



Swiss Institute of
Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

Overview of Single-Cell Transcriptomics Workflow Strategies

Dr. Nikolai Püllen, FGCZ

16.06.2026, Zurich



Introduction

The single-cell revolution

General overview of sequencing

NGS technology

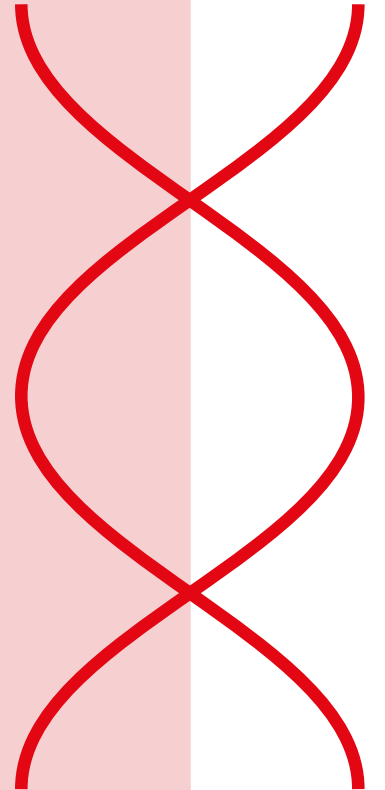
Long-read

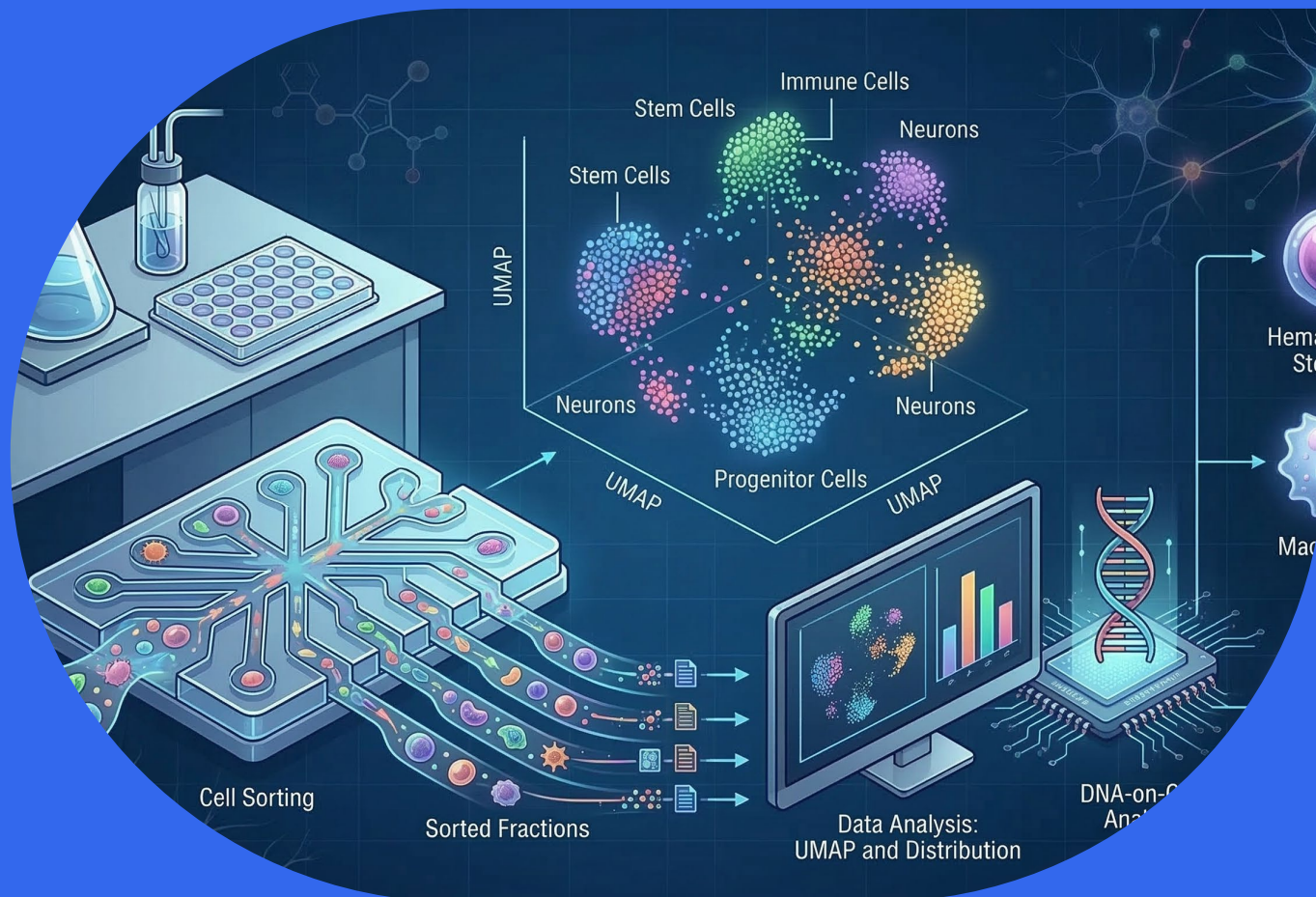
Single-Cell Transcriptomics

Technology overview

Major commercial technologies

Experimental design





AI-Generated by Google Gemini

Introduction

The Single-cell Revolution





A human is a mosaic of different cells

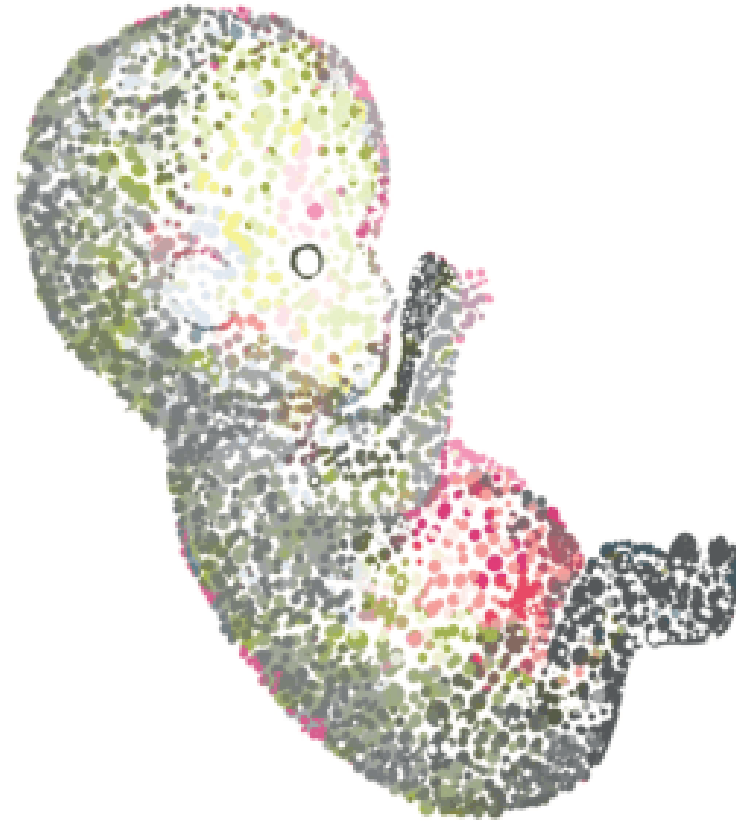
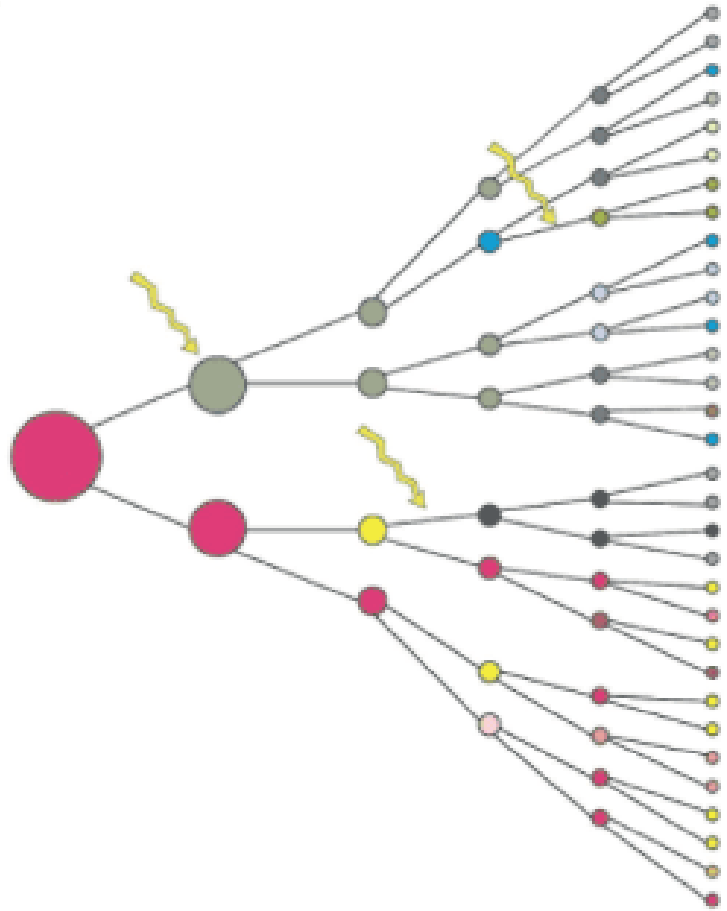
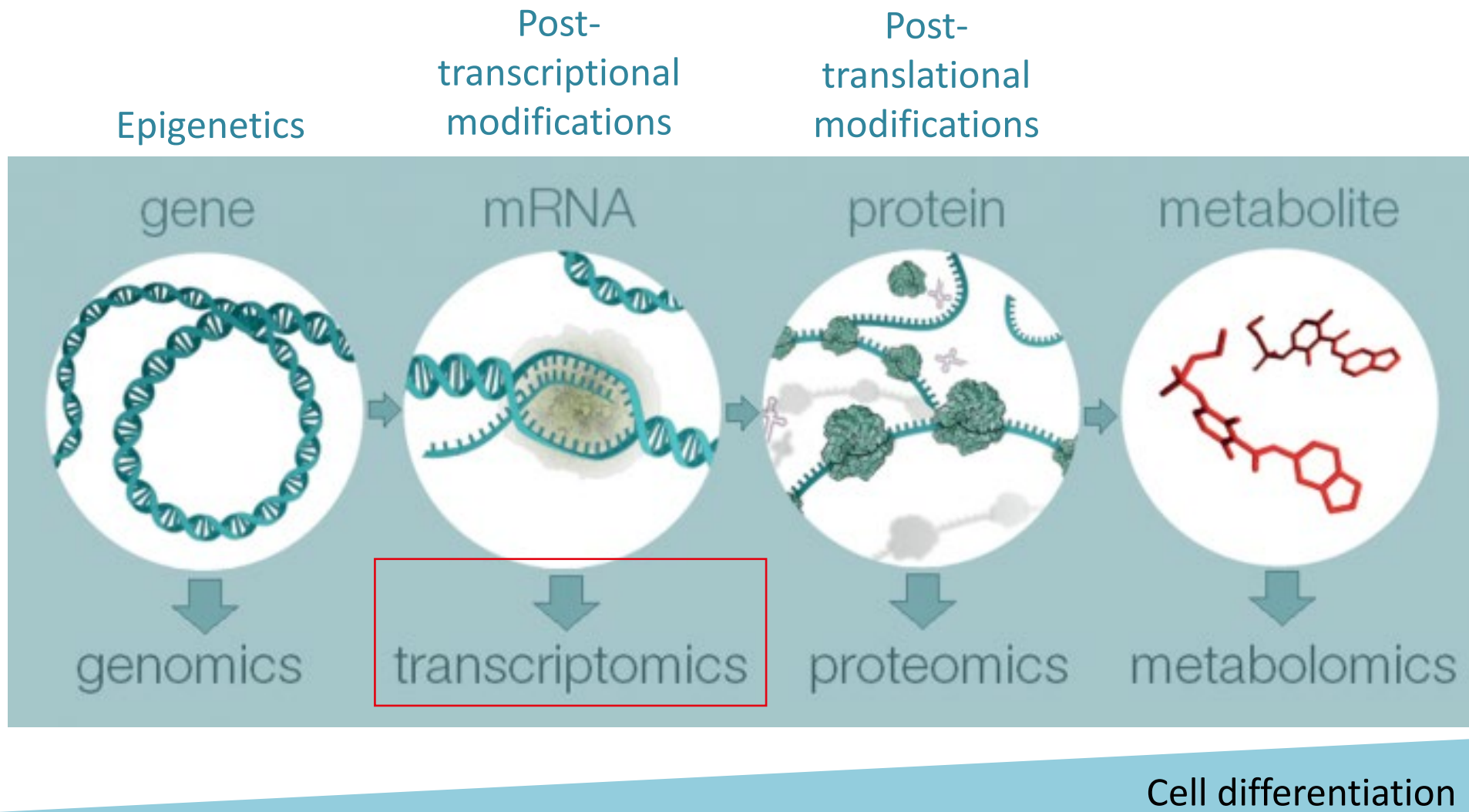


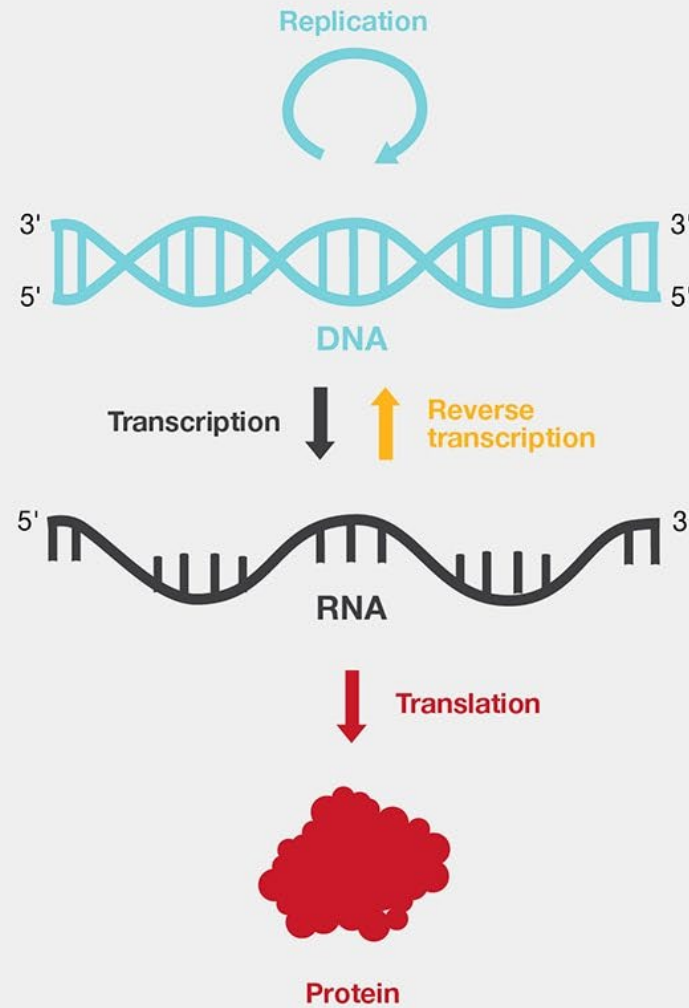
Illustration by Tarryn Porter

How can we characterize cell populations?





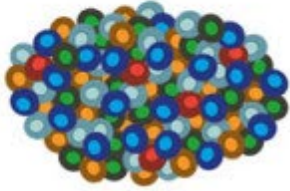
The updated dogma of molecular biology





Advantages of single-cell RNA-seq

Bulk tumor



Undetailed and incomplete picture

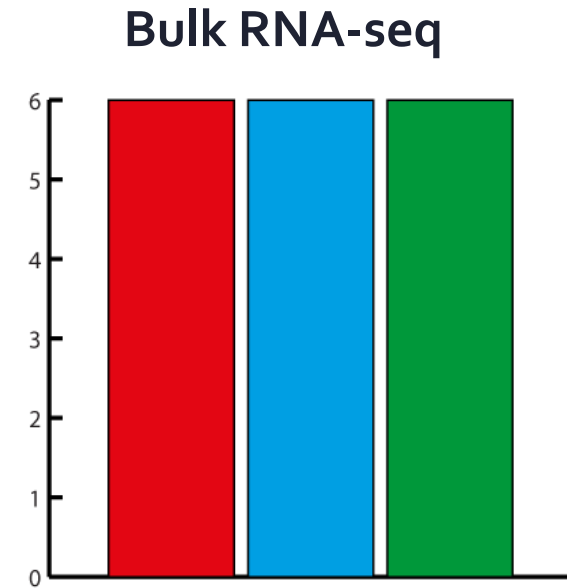
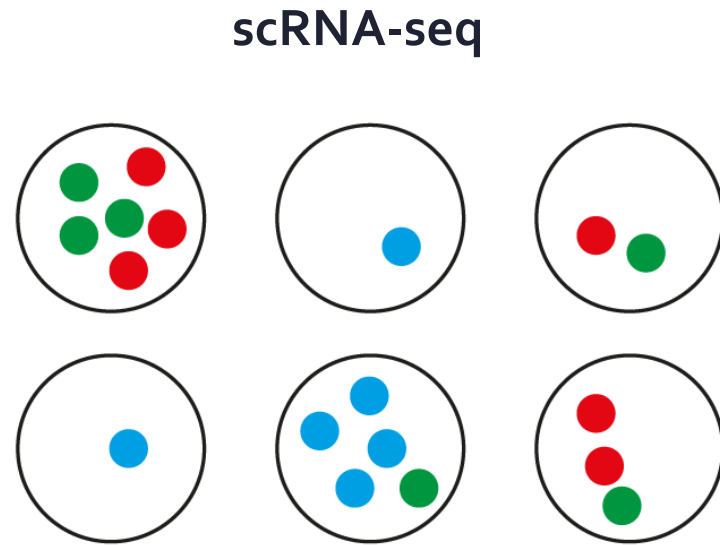


Inability to resolve cell populations





Advantages of single-cell RNA-seq

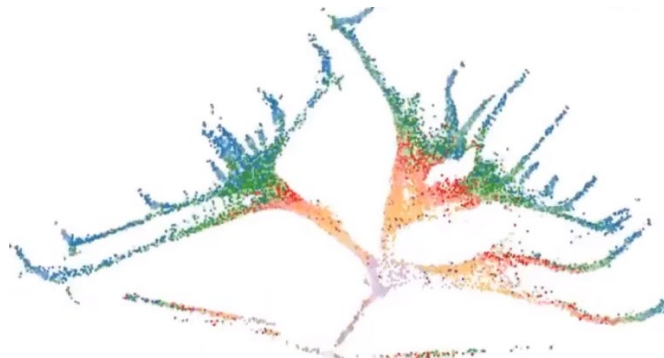


Adapted from Macaulay and Voet, Plos Genetics, 2014.



Power of single-cell transcriptomics technologies

"The single cell revolution is just starting"



2018
BREAKTHROUGH
of the YEAR

Science 361, 594–599 (2018)

Science

nature.com/nmeth

January 2020 Vol. 17 No. 1

nature methods



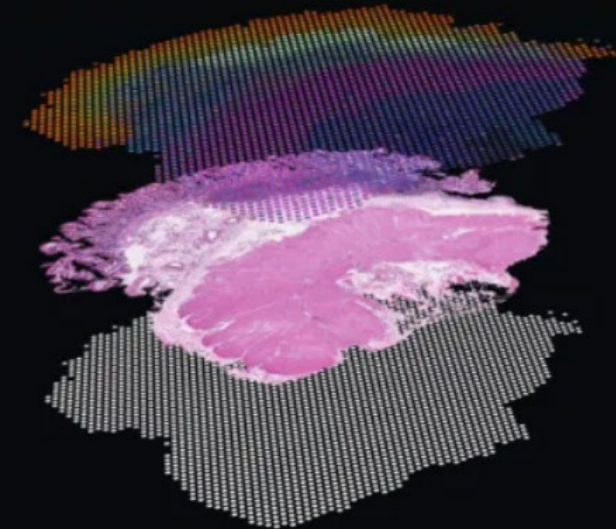
METHOD OF THE YEAR 2019

Localization microscopy twice as precise
A cryo-EM-based structural proteomics approach
Time-resolved crystallography at the European XFEL
Magnetic resonance at high speed

www.nature.com/nmeth / January 2021 Vol.18 No.1

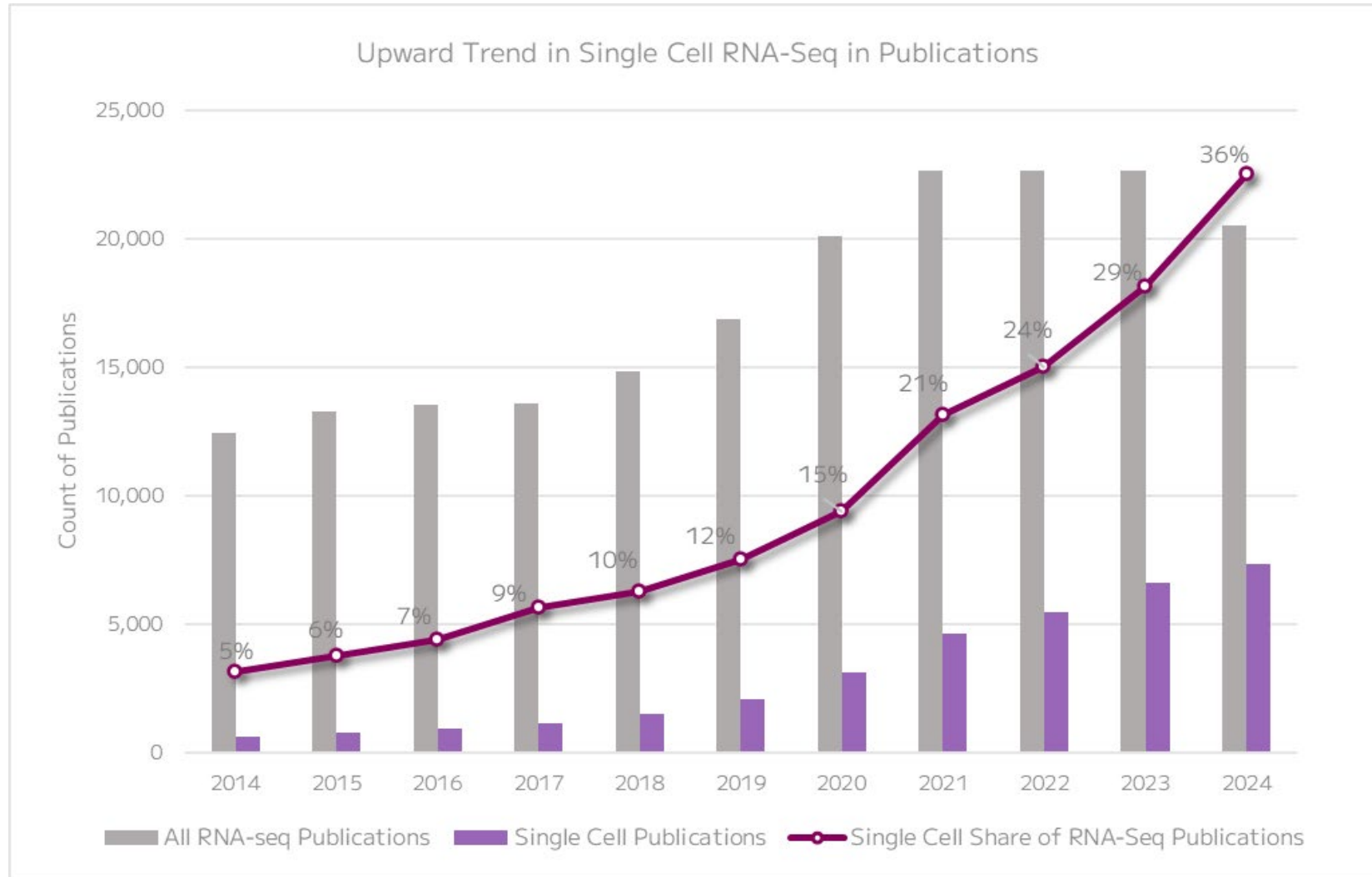
nature methods

Method of the Year 2020:
Spatially resolved transcriptomics





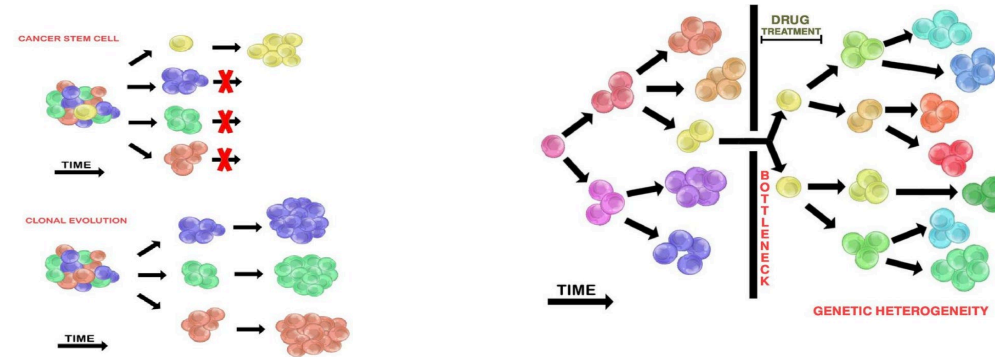
Single-cell RNA-seq studies over time



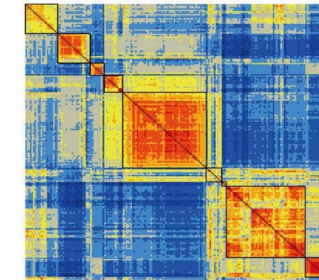
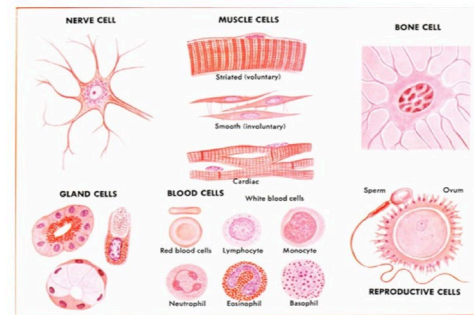


Main application of single-cell transcriptomics

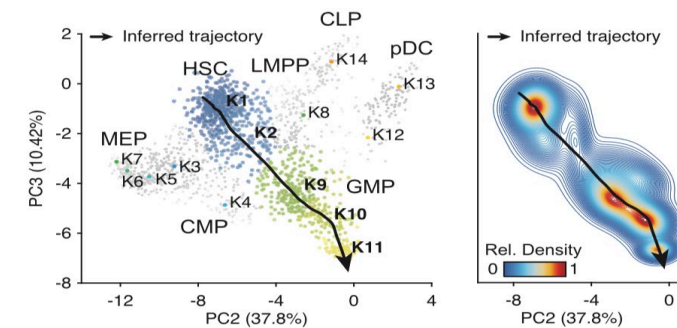
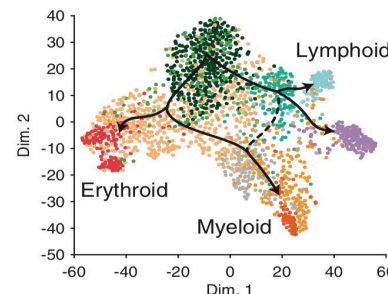
- Cell genetic and response heterogeneity



- Classification & identification of cells

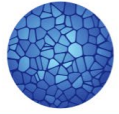


- Stem cell differentiation





The human cell atlas project



HUMAN
CELL
ATLAS

[Home](#) [HCA](#) [Areas of Impact](#) [News](#) [Publications](#) [Data Coordination](#) [Join HCA](#) [Contact](#)

MISSION

To create comprehensive reference maps of all human cells—the fundamental units of life—as a basis for both understanding human health and diagnosing, monitoring, and treating disease.

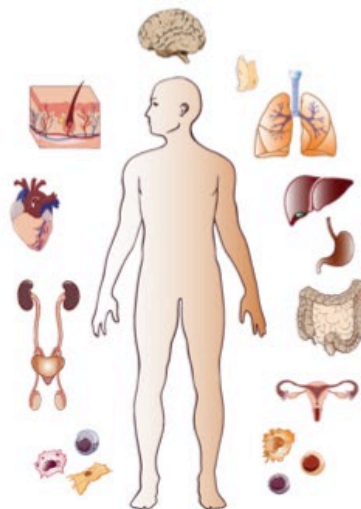
- Sequencing **all** the (37.2 trillion!) cells in the human body
- Sanger Institute, MIT and Harvard
- Mark Zuckerberg and Priscilla Chan donated \$3 billion for the project
- Started in 2016, first draft for release in **2026**





Potential impact of atlasing initiatives

A. Atlased organisms



Homo sapiens - adult and developmental



Malaria parasite
Plasmodium berghei



Caenorhabditis elegans



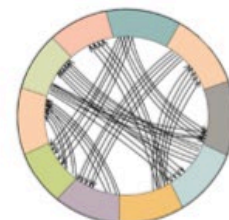
Danio rerio

B. Potential impact



Identification of novel cell types

Understanding of chromatin states



Signalling networks

Cellular interactions

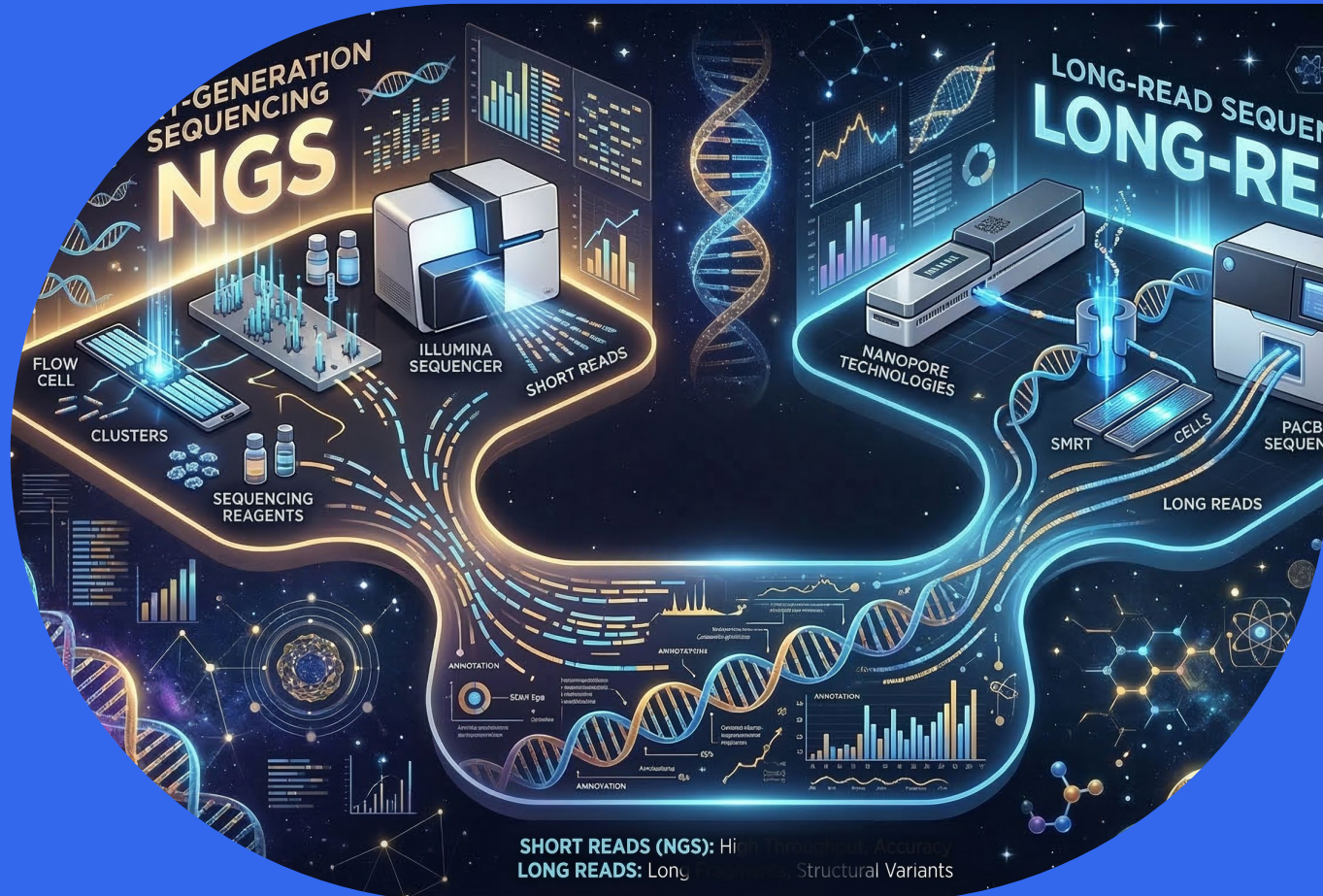


Tissue organisation

Drug discovery and medicine



Improvements in regenerative medicine and *in vitro* models



AI-Generated by Google Gemini

Sequencing Technology Overview

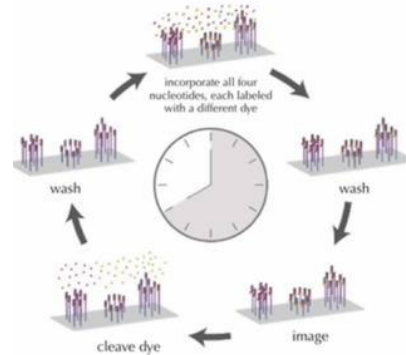
Next Generation and Long-Read Sequencing



Short read vs. long read technology

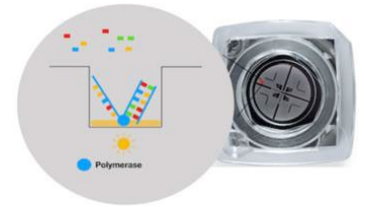
2nd Generation

- Monoclonal amplification on a solid surface
- Sequencing in cycles
- **Short** reads
- **High number** of reads



3rd Generation

- No amplification
- Sequencing on single molecules
- **Long** reads
- **Low number** of reads



illumina[®]

E Element
Biosciences

UG ULTIMA
GENOMICS



PacBio

Oxford
NANOPORE
Technologies

SIB

Illumina sequencers

Focused Power



iSeq™ 100



MiniSeq™



MiSeq™

Flexible Power



NextSeq™ 550



NextSeq™ 2000

Production Power



NextSeq™ 2000



HiSeq™ 4000

Population Power



HiSeq™ X



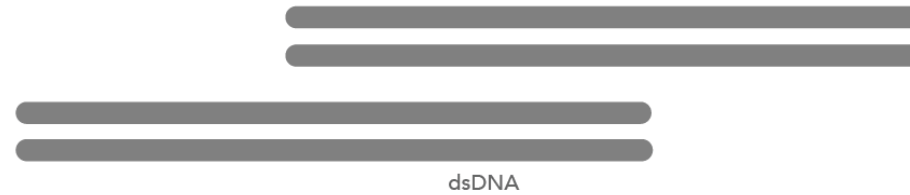
NovaSeq™ 6000



NovaSeq X

Standard short-read DNA library preparation

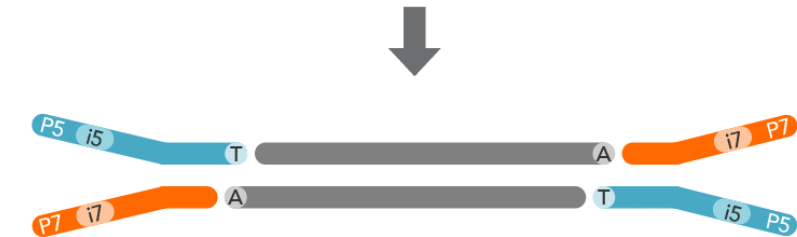
Fragmentation



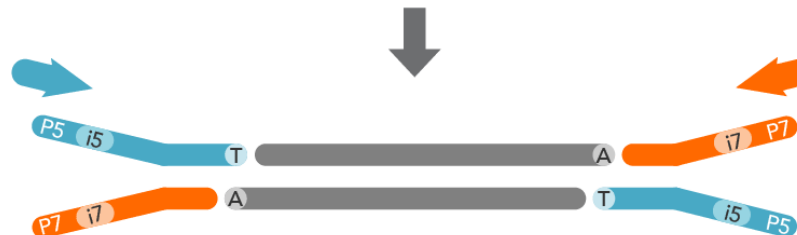
End repair and A-tailing



Ligation

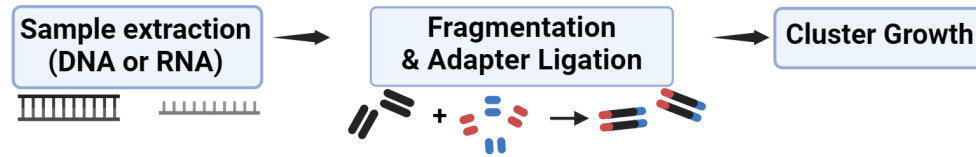


PCR amplification





Illumina sequencing method



High density cluster formation

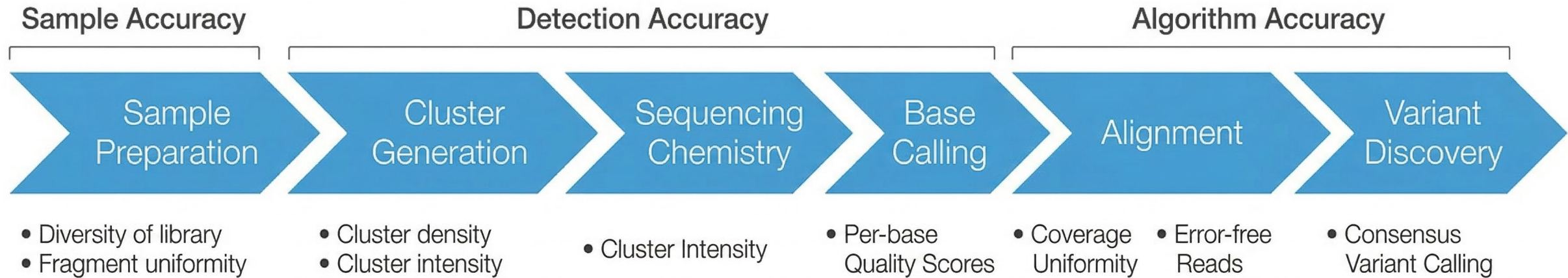
- 1) Fragments annealing to flow cell
- 2) Reverse strand synthesis
- 3) Dissociation
- 4) Bridge formation
- 5) Bridge amplification
- 6) -Repeat-
- 7) Dissociation

Sequencing by synthesis

- 1) Denaturation
- 2) Incorporation
- 3) Imaging
- 4) Cleavage
- 5) -Repeat-



Short-read workflow

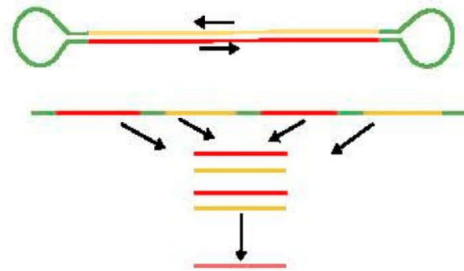


Factors that Contribute to Platform Accuracy



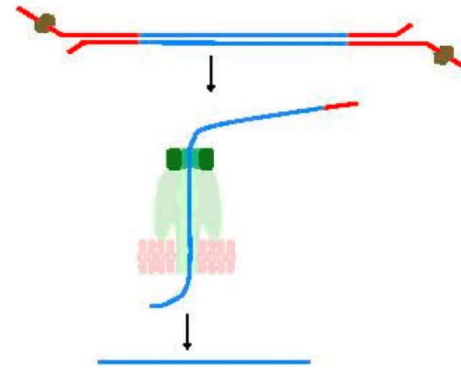
Long-read sequencing

PacBio

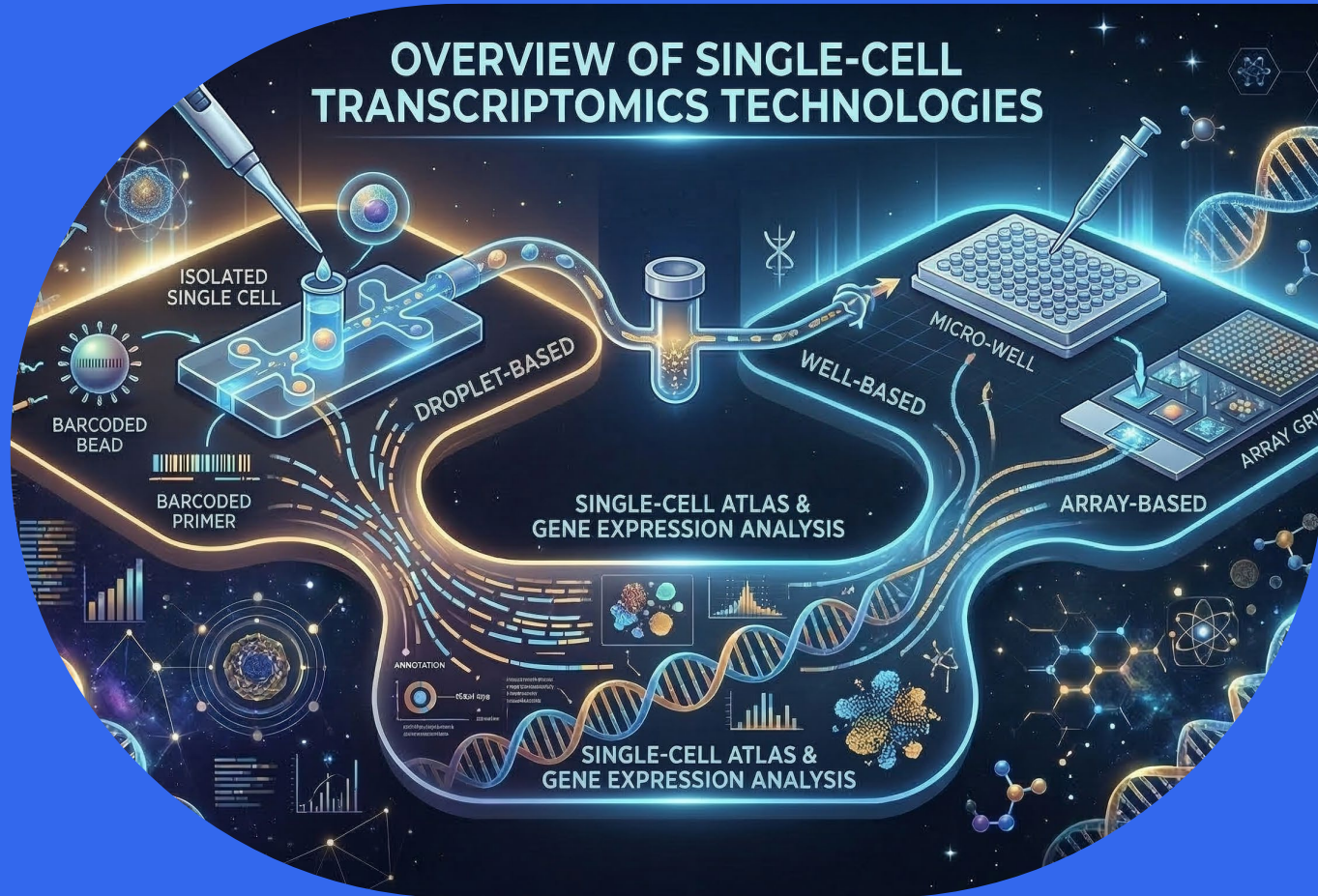


- Long-read lengths

Oxford Nanopore



- Ultra-long read lengths



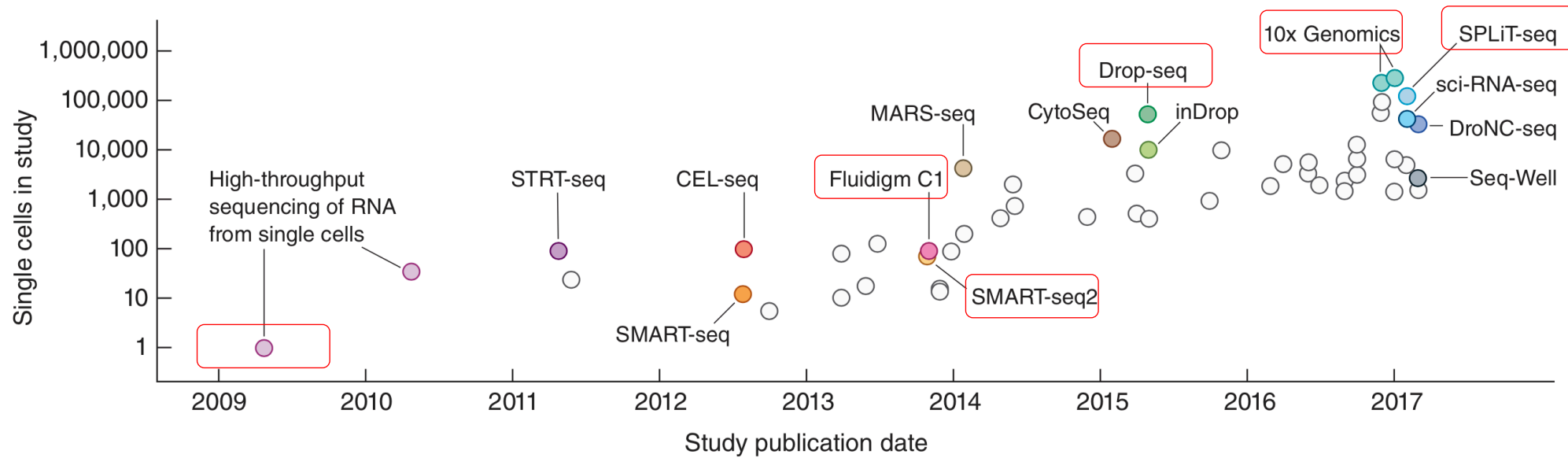
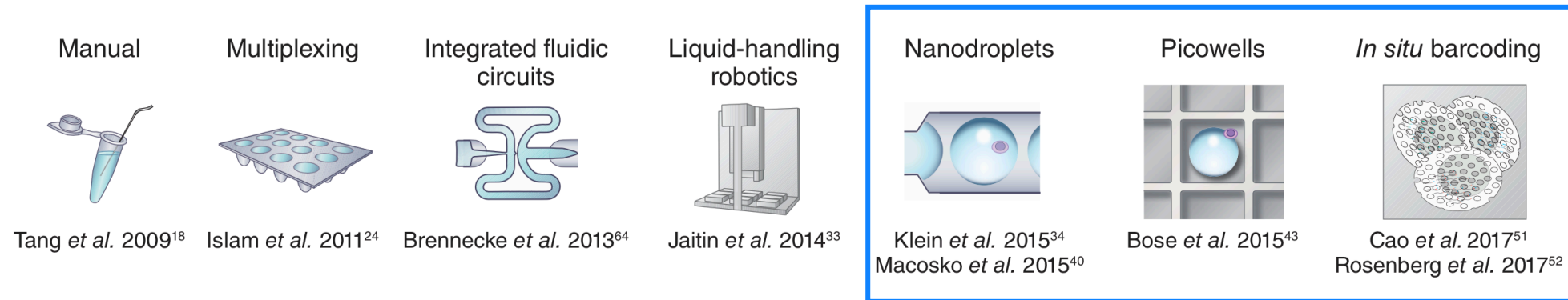
AI-Generated by Google Gemini

Single-Cell Transcriptomics

Single-Cell Technology Overview



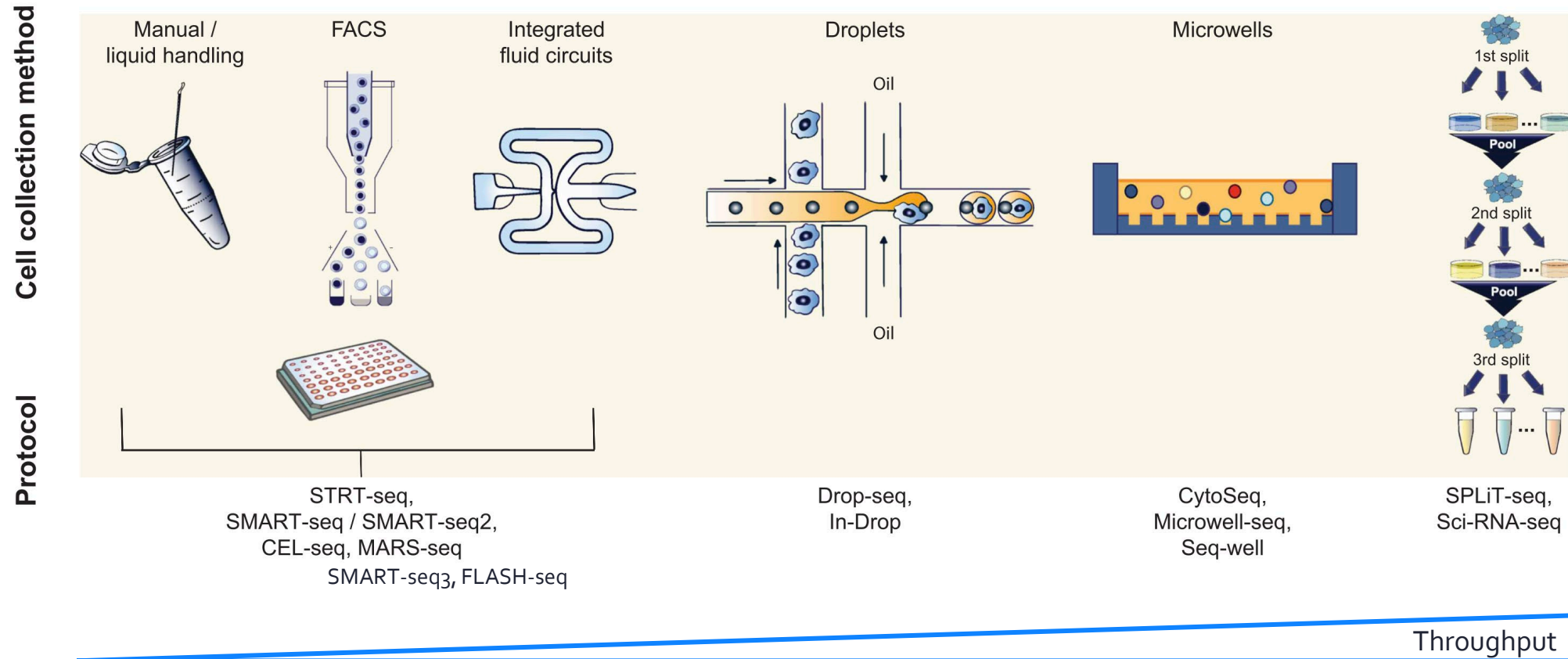
Increasing throughput of single-cell RNA-seq



Valentine S. *et al.*, *Nature protocol*, 2018.

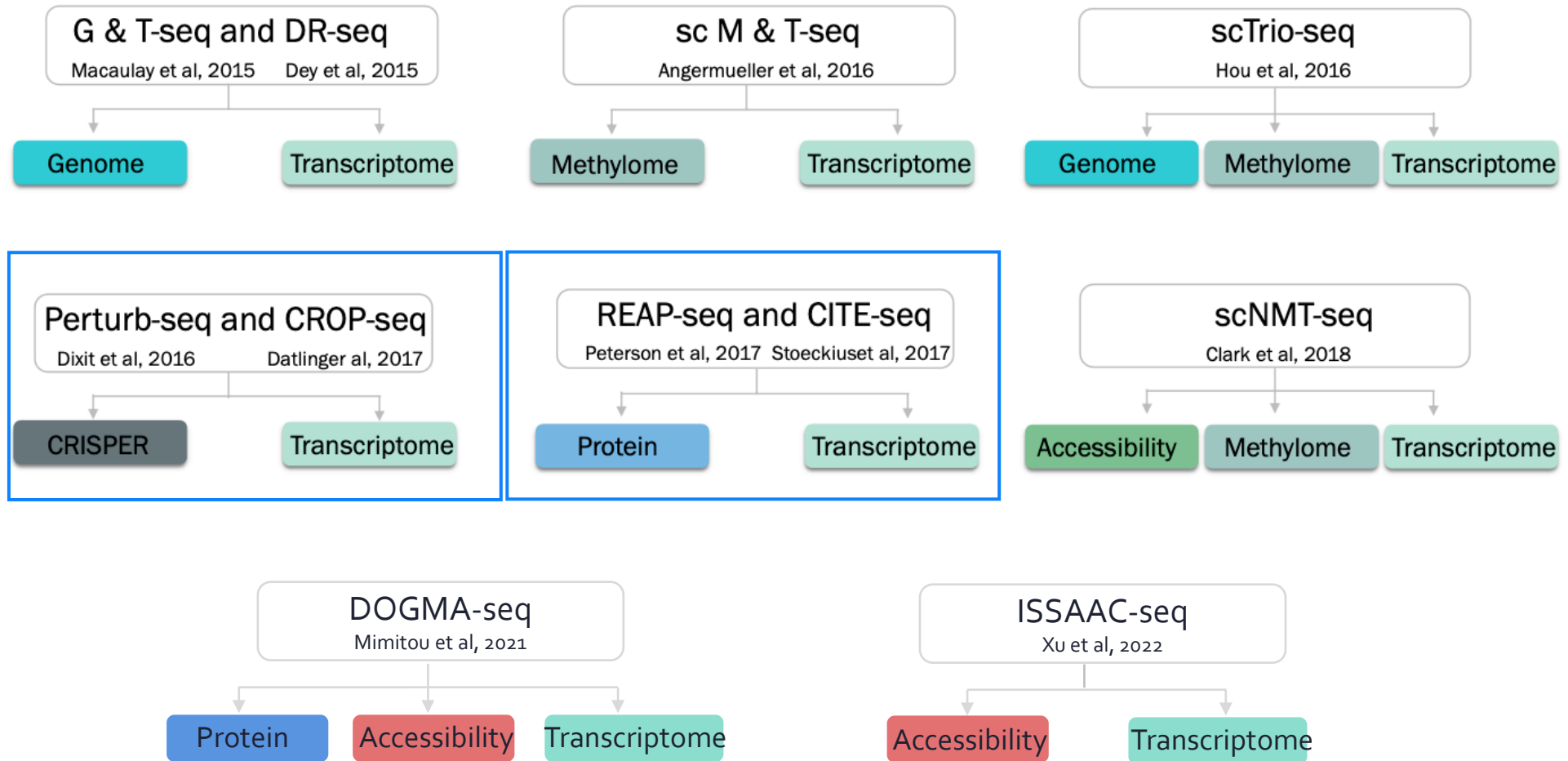


Current single-cell RNA-seq methods





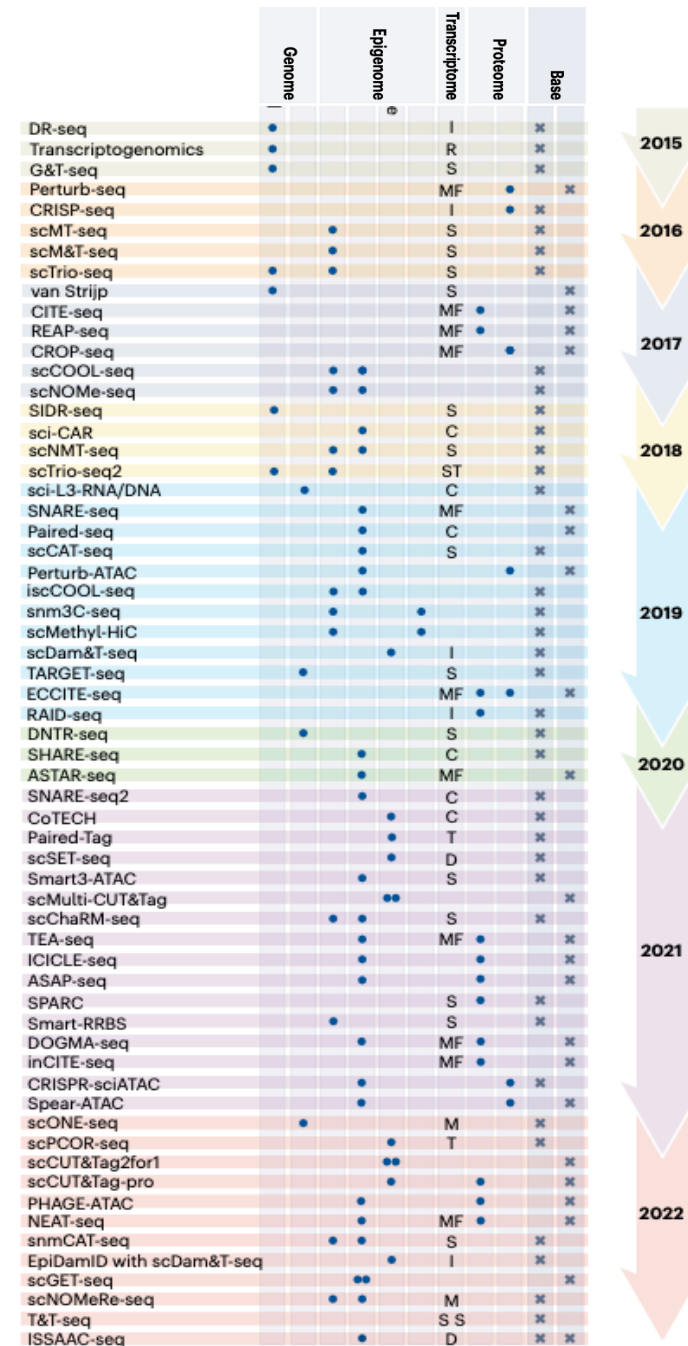
Single-cell multiome methods





Single-cell multiome methods

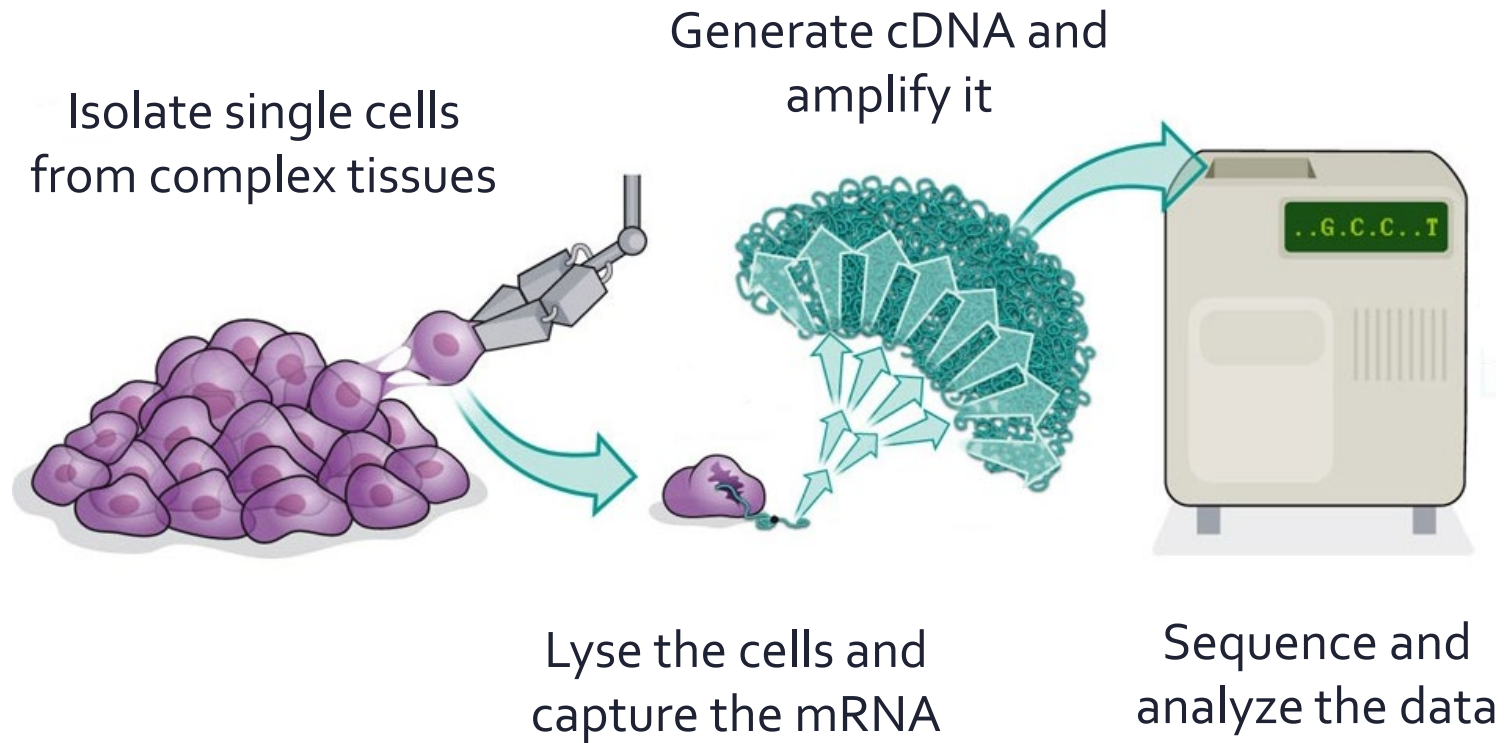
- Huge variety of methods available
- Continuously increasing
- Different targets
- Many not commercialized



Single-cell and Spatial multi-omics, Vandereyken et al., 2023.



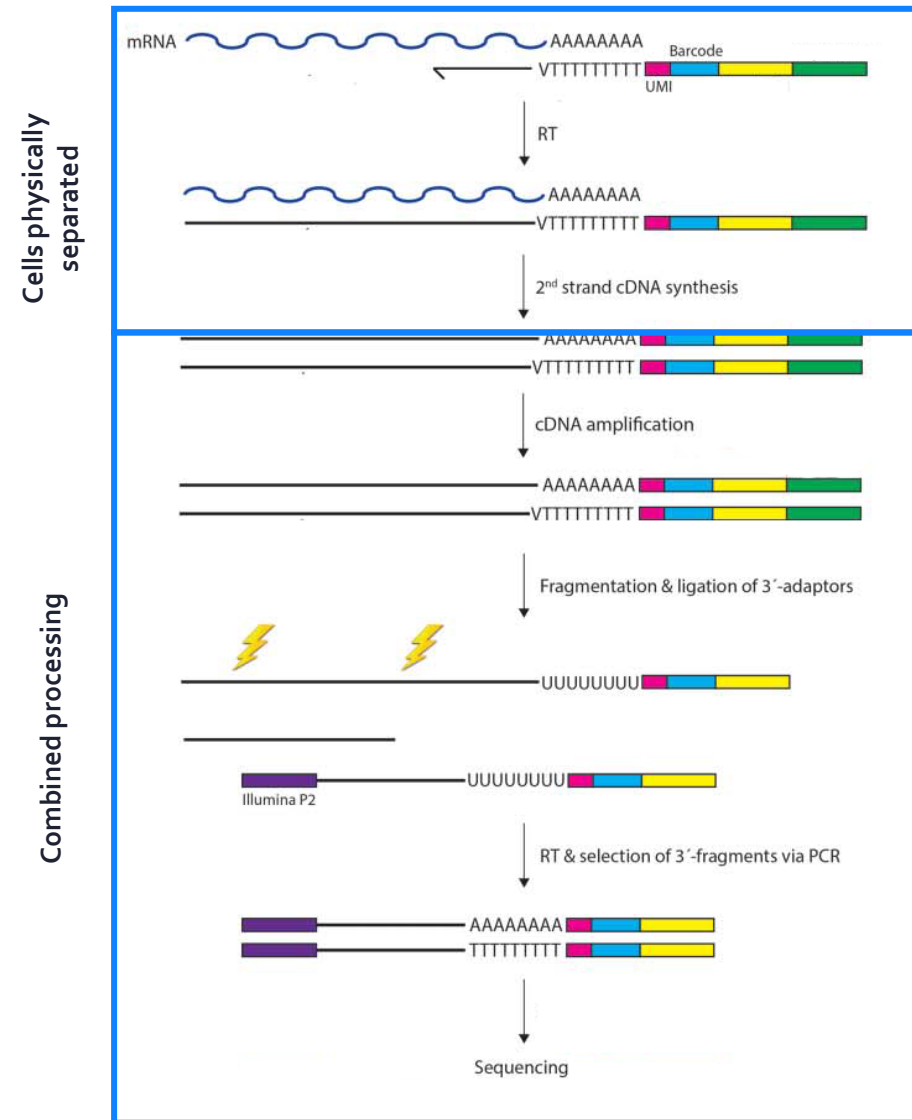
How does single-cell RNA-seq work?





RT-based approach

- ✓ Captures actual RNA
- ✓ High-throughput
- 3' or 5' coverage only
- Less sensitive compared to probes



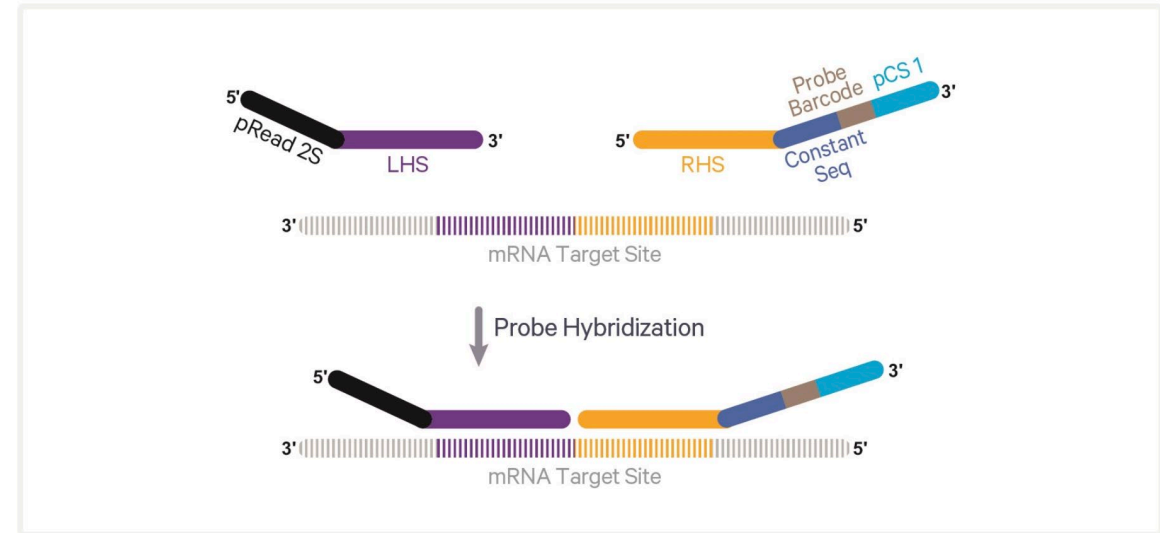
Adapted from Picelli, RNA Biology, 2017.



Probe-based approach

- ✓ Works with fixed sample
- ✓ Lower price compared to RT
- ✓ Very high throughput and sensitivity
- Sequencing probes instead of RNA
- Organism probeset required
- High cell number required

Probe Hybridization



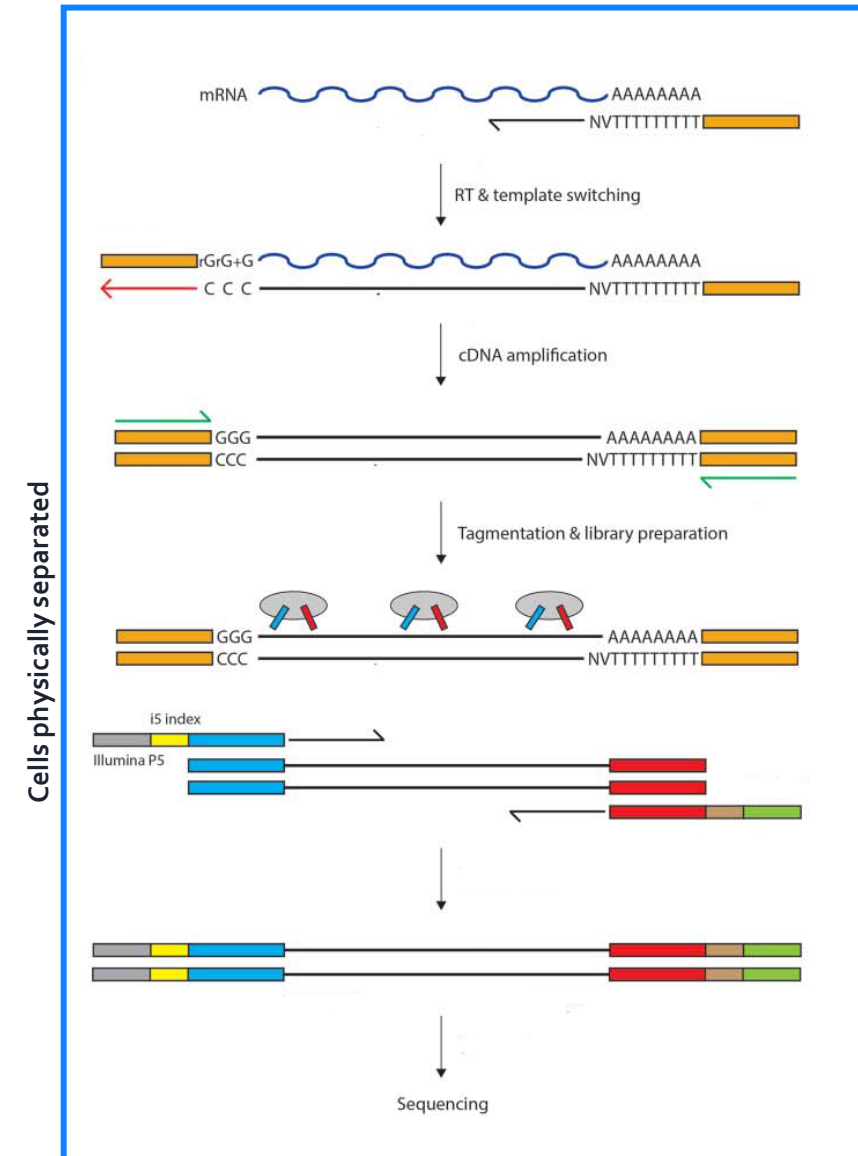
Probe summary: Human WTA Probeset

- >18,000 coding genes specifically detected
- >55,000 probe pairs
- Design strategy: maximize **specificity, sensitivity and sequencing efficiency**



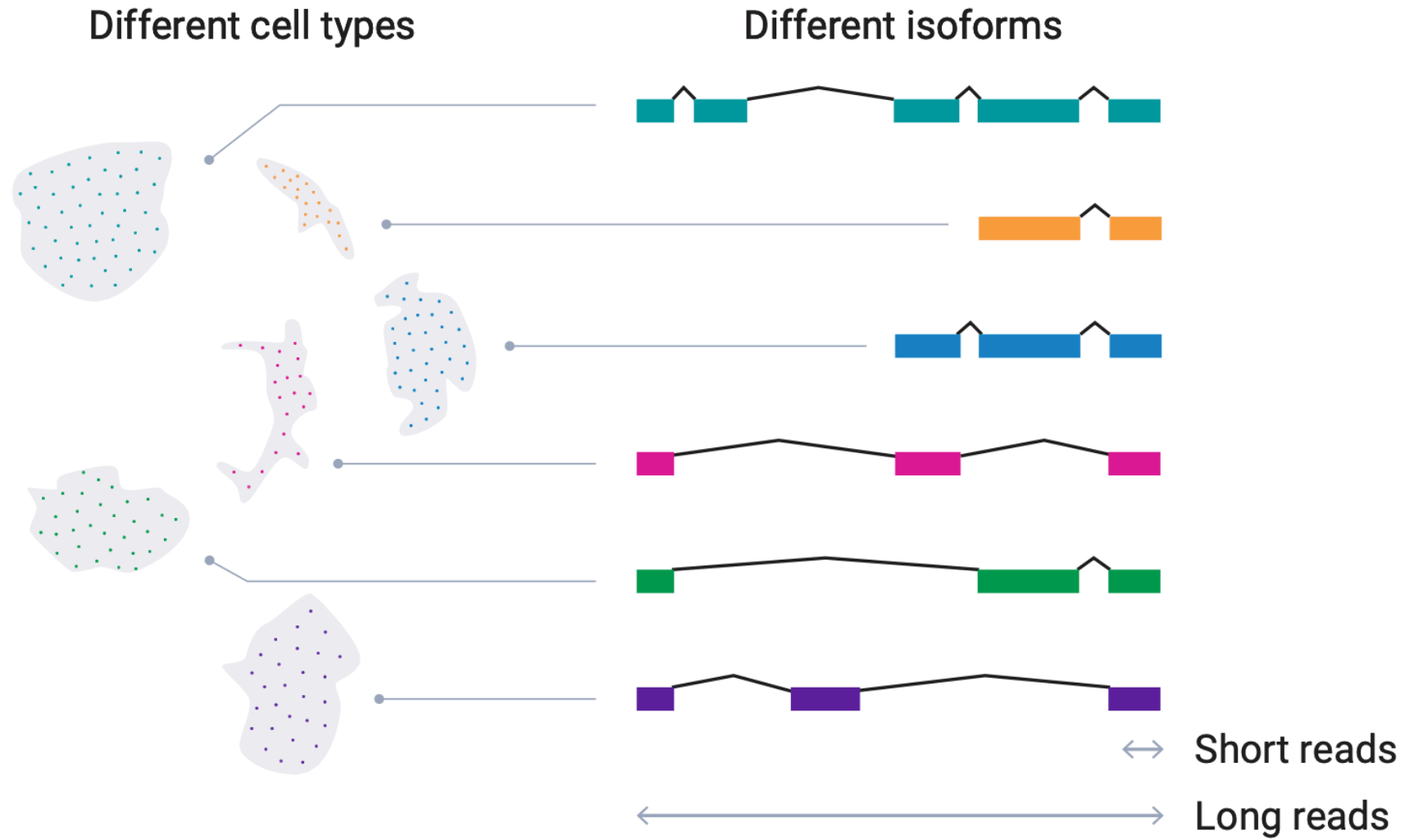
Single-cell RNA-seq methods: full-length methods

- ✓ Provides full length coverage of the transcripts
- ✓ High sensitivity
- Low throughput
- Expensive



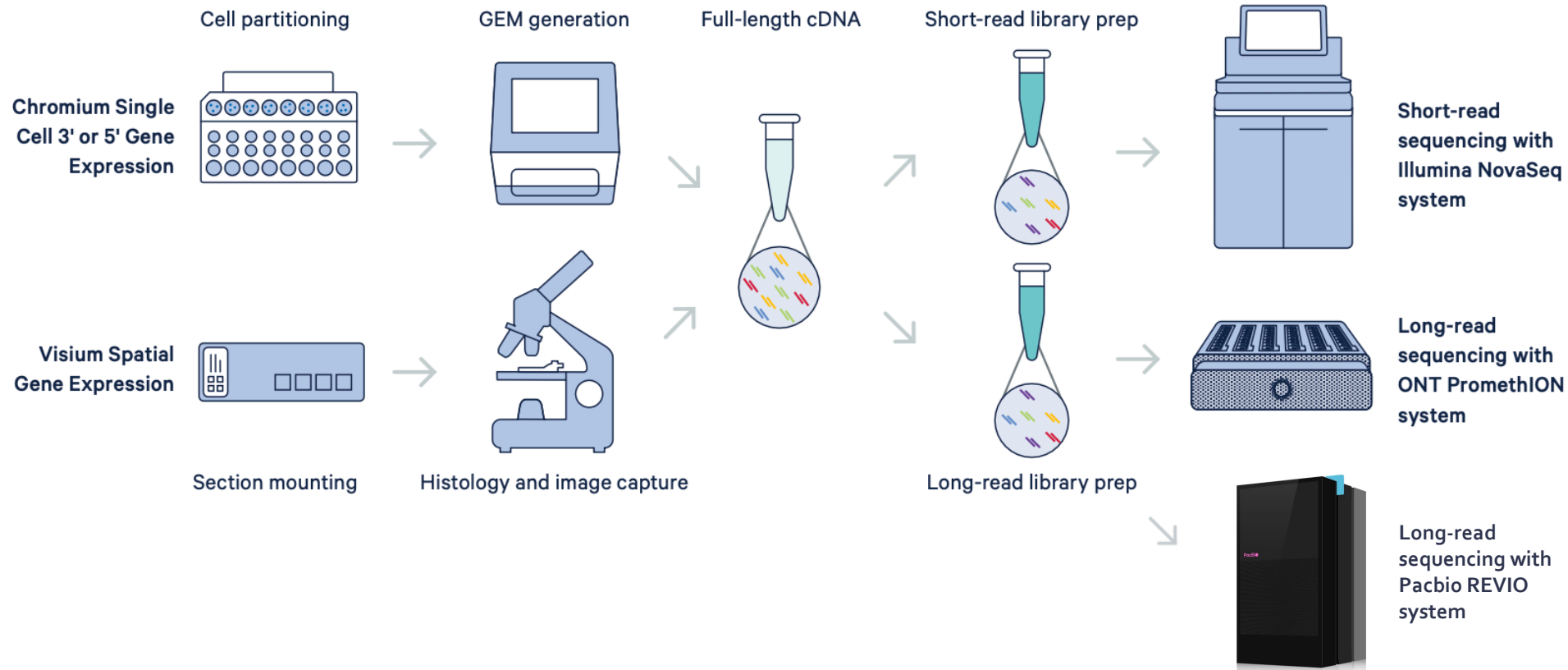
Adapted from Picelli, RNA Biology, 2017.

Single-cell long-read RNA-seq methods





Combining single-cell with long-read sequencing





AI-Generated by Google Gemini

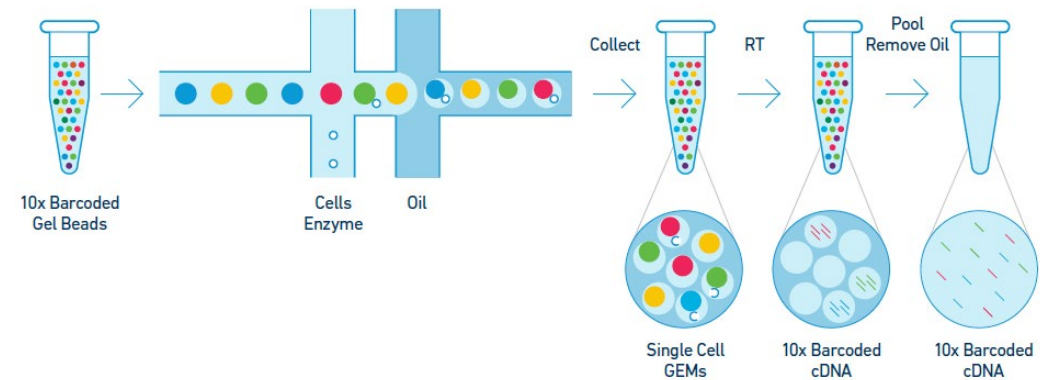
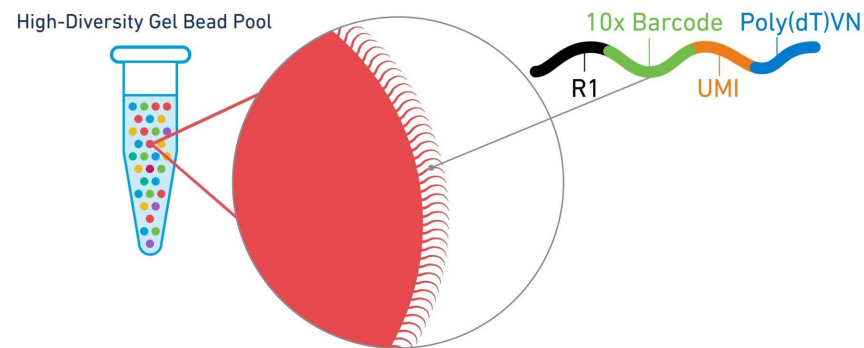
Single-Cell Transcriptomics

Major Commercial Technologies



Droplet based Platform: 10x Genomics

- ✓ Current standard in the field
- ✓ Simple workflow
- ✓ Multiomic applications
- Cell size $< 50 \mu\text{m}$
- Clean samples crucial





10x scRNA-seq assay overview

Assay types

Universal 3' Gene Expression
Universal 5' Gene Expression
Epi Multiome ATAC + Gene Expression
Flex Gene Expression

→ Gene expression
→ Immune repertoire analysis
→ scATAC-seq
→ Fixed cells

Protocol extensions

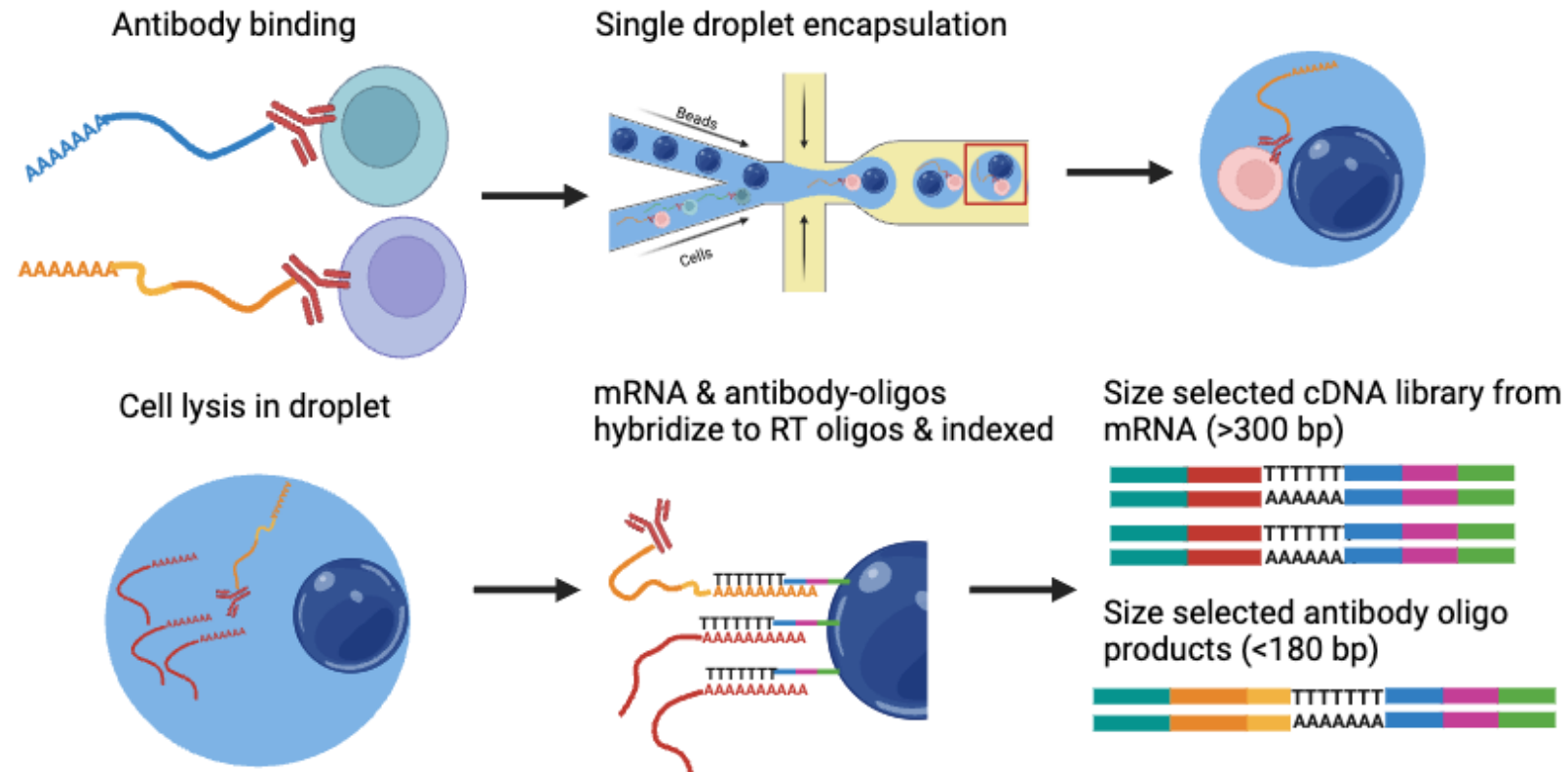
- » On Chip Multiplexing
 - » Cell-surface proteins
 - » Intracellular proteins
 - » BCR/TCR
 - » CRISPR Screen
 - » Long-read
- } **CITE-Seq**



10x Genomics, 2022



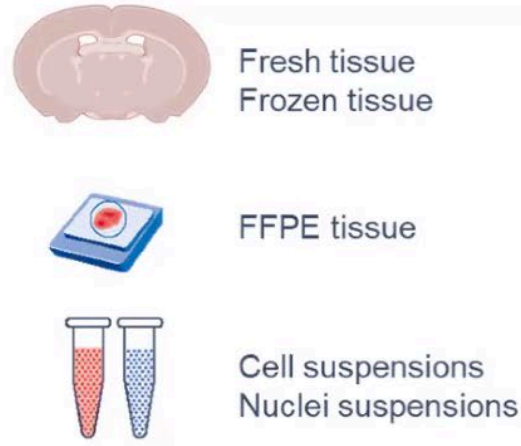
CITE-Seq with 10x Genomics



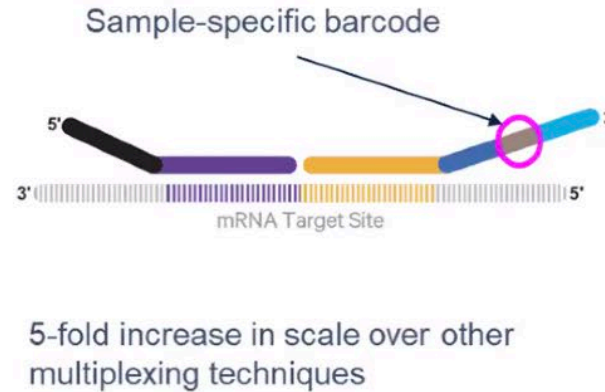


10x Gene Expression Flex (Apex)

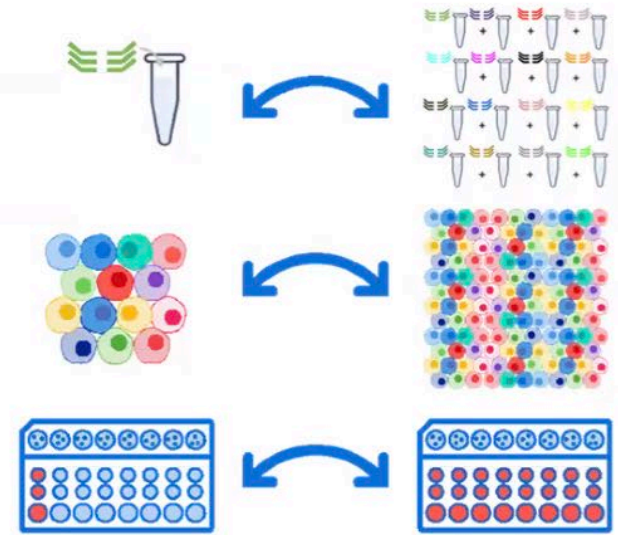
Broad sample compatibility



In-line multiplexing



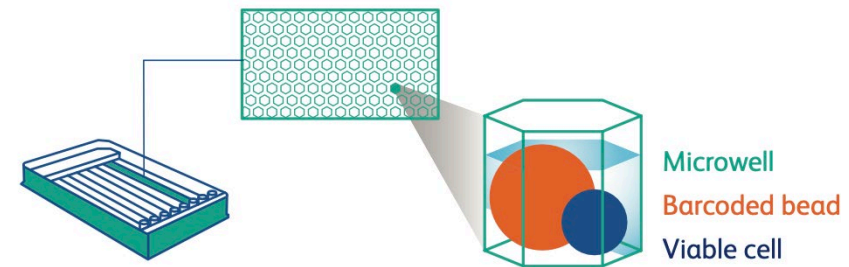
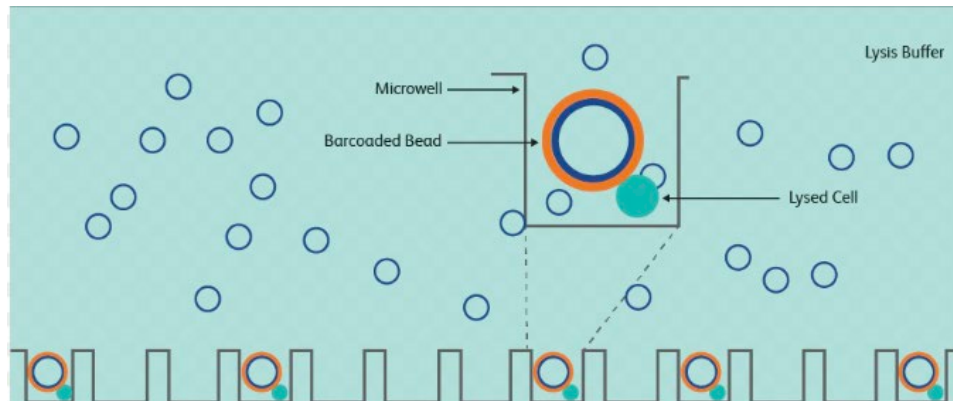
Flexible sample and cell input number



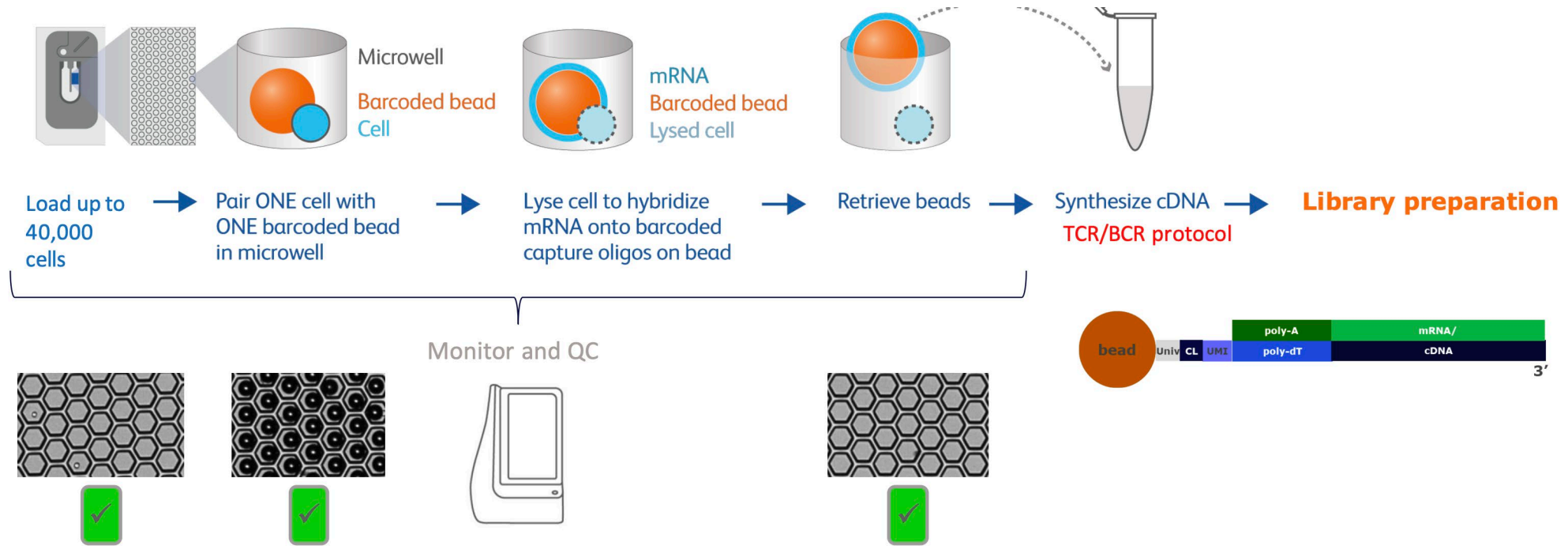
- ✓ Sample are fixed
- ✓ Handling various types of samples (F,FF,FFPE)
- ✓ More flexible for sample multiplexing
- Only available for Human and Mouse
- High number of cells required

Microwell-based platform: BD Rhapsody HT

- ✓ Visualization of cell capture
- ✓ Less sensitive for unclean samples
- ✓ Lower multiplet rate than 10x
- ✓ Multiomic analysis
- Lengthy library preparation protocols
- Cell recovery lower than 10x



Microwell-based platform: BD Rhapsody HT





BD scRNA-seq assay overview

Assay types

WTA Gene Expression

Targeted Gene Expression

VDJ-Seq

ATAC-Seq + Gene Expression

→ Global gene expression

→ Selected gene expression

→ Immune repertoire analysis

→ scATAC-seq

Protocol extensions

❖ (Nuclei) Multiplexing

❖ Omics Guard

❖ Cell-surface proteins

❖ Intracellular proteins

❖ BCR/TCR

❖ CRISPR Screen

❖ Long-read

CITE-Seq





Instrument-free scRNA-seq: Parse bio

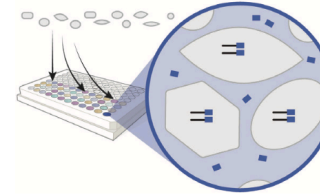
- ✓ Fixed samples
- ✓ Sample multiplexing
- ✓ Super high-throughput
- Complicated and lengthy workflow
- Low cell recovery rate

Semi-automation possible



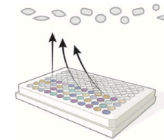
1 Reverse Transcription

Split | Fixed cells/nuclei are distributed into wells, and the first sample-specific barcodes are added by in-cell reverse transcription.



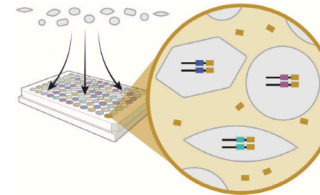
2 Pool

Pool | All the cells are pooled together.



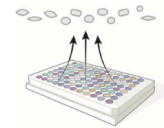
3 Ligation

Split | The pooled cells are distributed across a plate, and an in-cell ligation adds the second barcode.



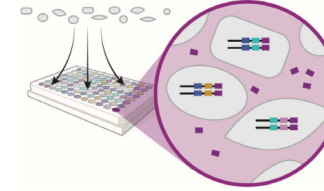
4 Pool

Pool | All the cells are pooled together.



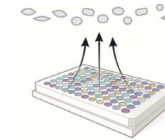
5 Ligation

Split | The pooled cells are again distributed across a plate, and a third barcode is added via in-cell ligation reaction.



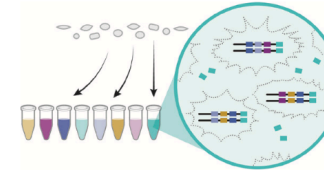
6 Pool

Pool | All the cells are pooled together.



7 Lysis and Library Prep

Split | The pooled cells are distributed across several sublibraries then lysed. The fourth barcode is added via PCR.



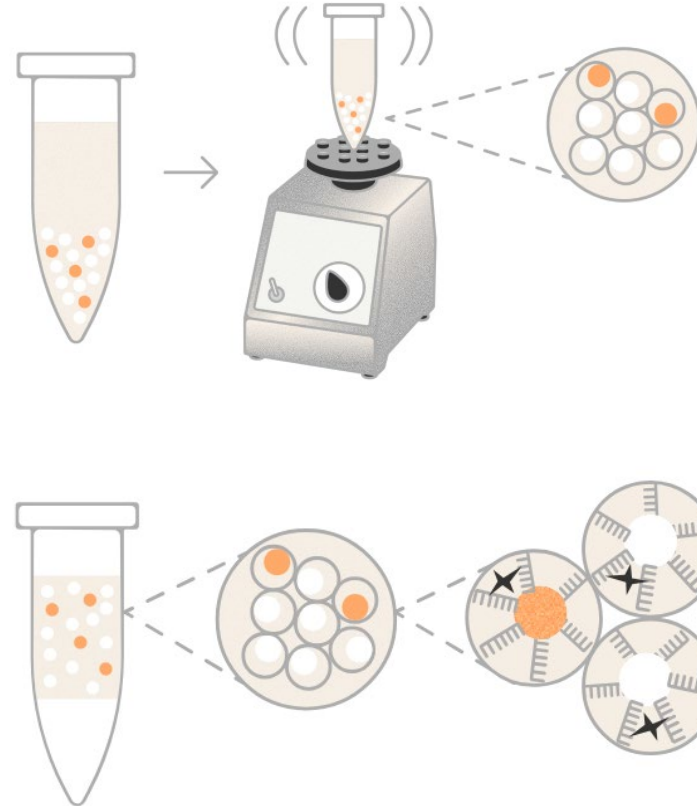
8 Sequencing and Analysis

Each transcript is assigned to a single cell based on a unique combination of barcodes.

Genes	Barcodes				
	1	2	3	4	
Gene A	—	—	—	—	Cell 1
Gene B	—	—	—	—	
Gene C	—	—	—	—	
Gene A	—	—	—	—	Cell 2
Gene B	—	—	—	—	
Gene D	—	—	—	—	
Gene E	—	—	—	—	Cell 3
Gene F	—	—	—	—	
Gene G	—	—	—	—	

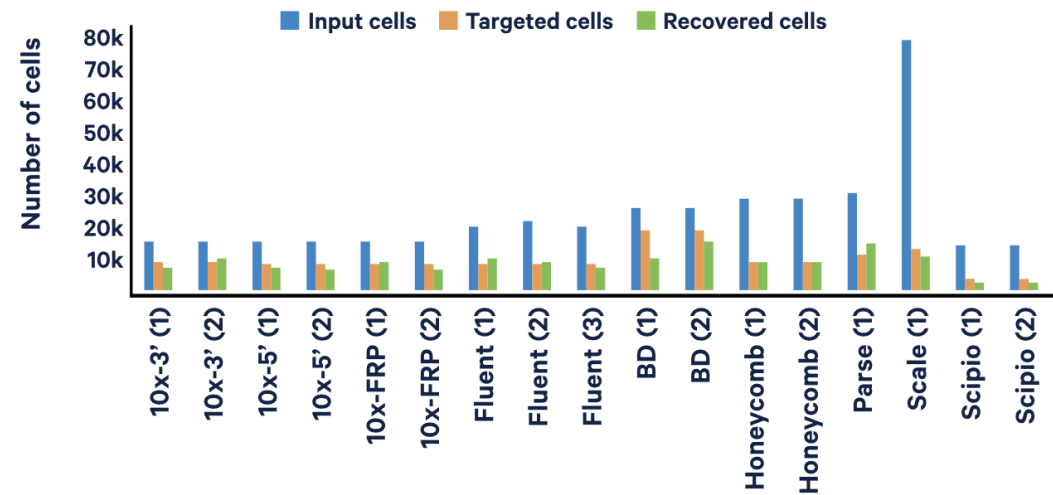
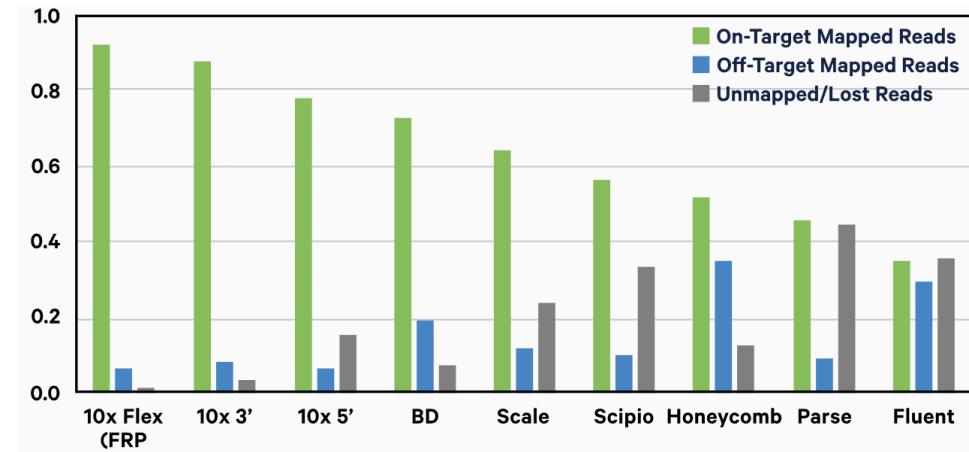
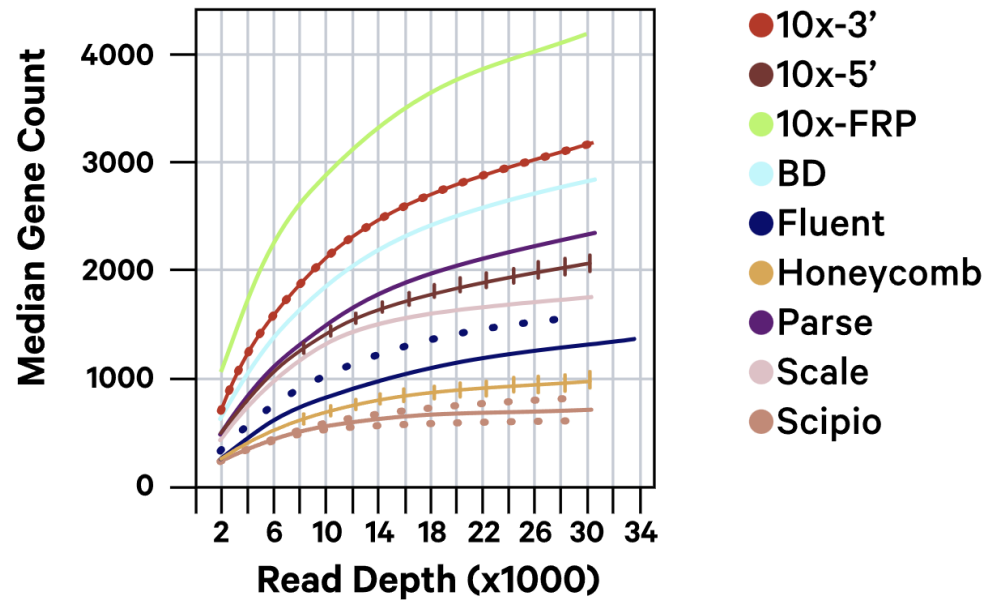
Instrument-free scRNA-seq: Illumina PIP-Seq

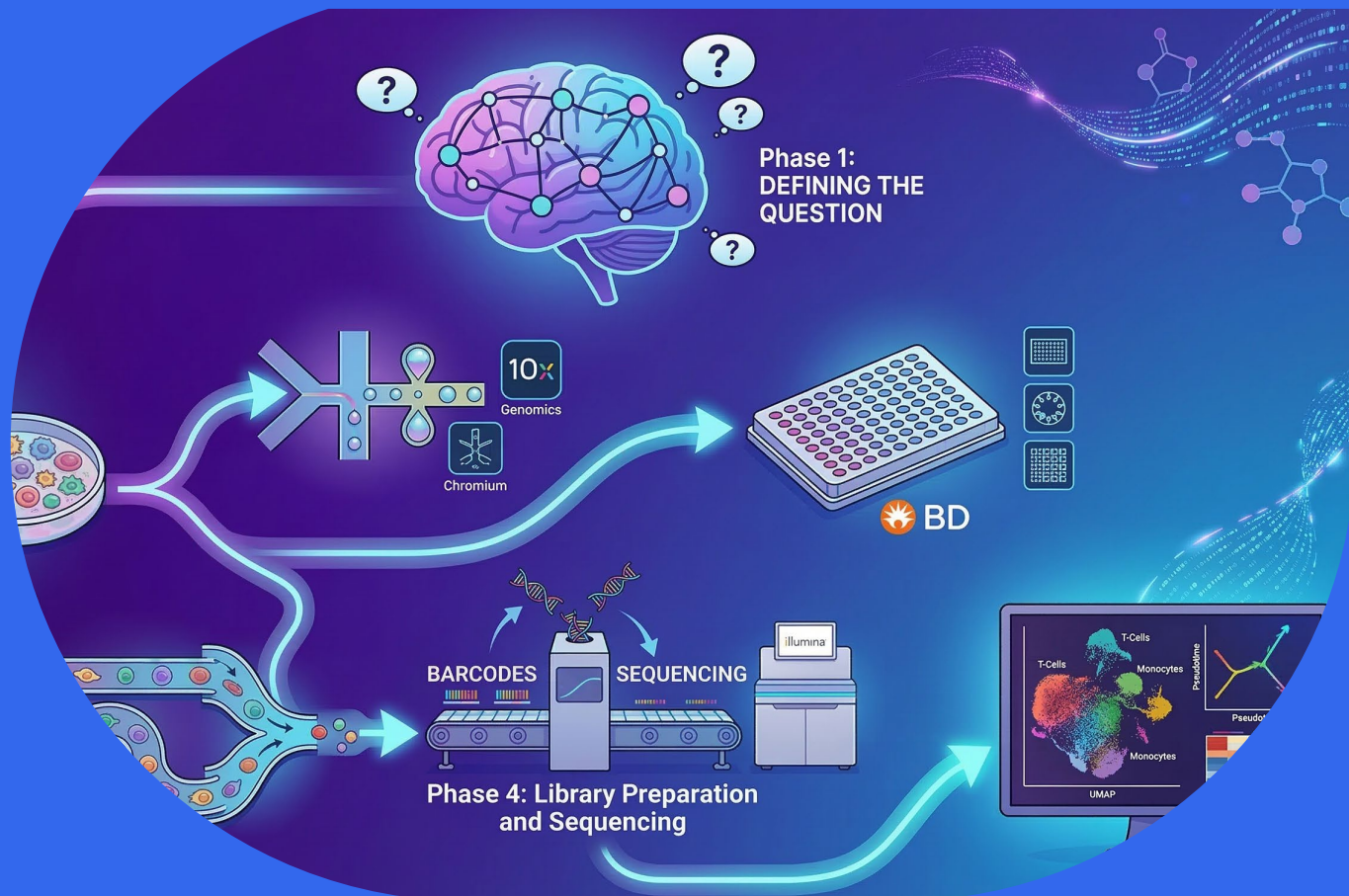
- ✓ Fresh or MeOH-fixed samples
- ✓ Low cost and hands-on time
- ✓ Any (poly-A) species
- Lower sensitivity compared to 10x/BD
- Max. 8 samples / 100k cells
- Low cell recovery rate





scRNA-seq methods comparison





AI-Generated by Google Gemini

Single-Cell Transcriptomics

Experimental Design

WHAT IS YOUR RESEARCH QUESTION?





Some common questions

Which technology?

→ Sample type, budget, logistics, throughput

How many replicates?

→ More

How many cells?

→ What populations are you looking for?

How many reads?

→ What genes are you looking for?



Finding the equilibrium





Most common pitfalls

1) **Complexity too high**

→ Increasing quantity usually decreases quality

2) **Technology does not fit sample type**

→ Choose wisely, know the parameters

3) **scRNA-seq is not necessary**

→ Do you expect different populations?

4) **Pilot experiment skipped**

→ Don't!



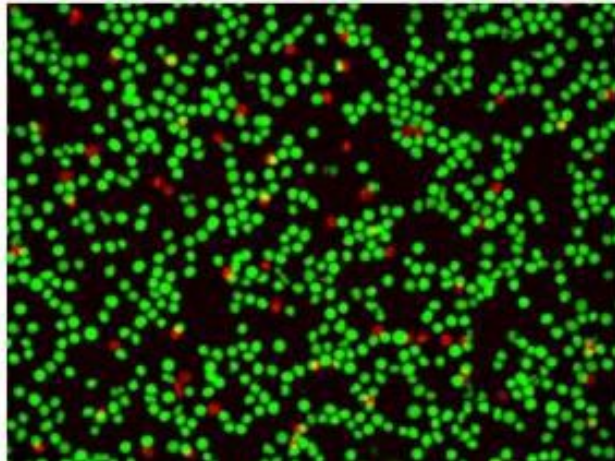
scRNA-seq requirements

High quality single-cell suspension

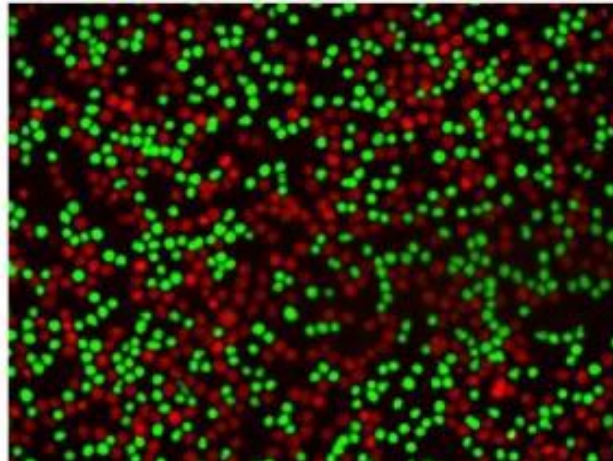
- » No clumps
- » Debris-free
- » Sufficient cell number
- » High viability

Viability:

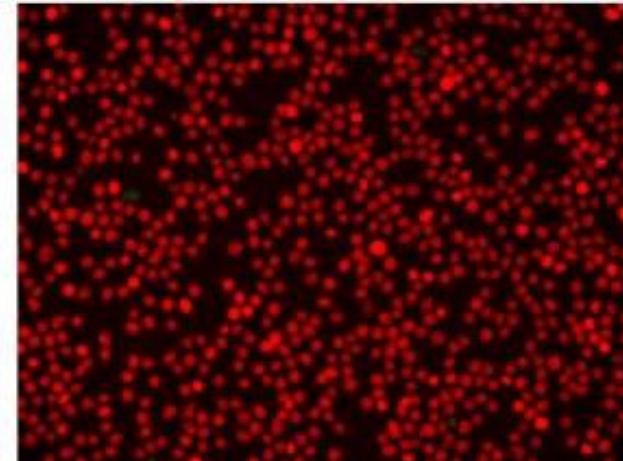
90%



50%

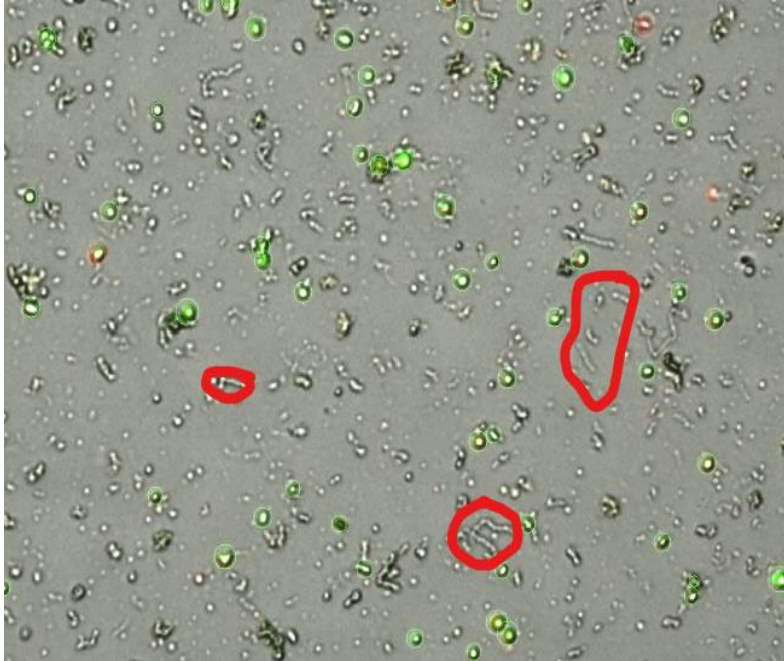


0%

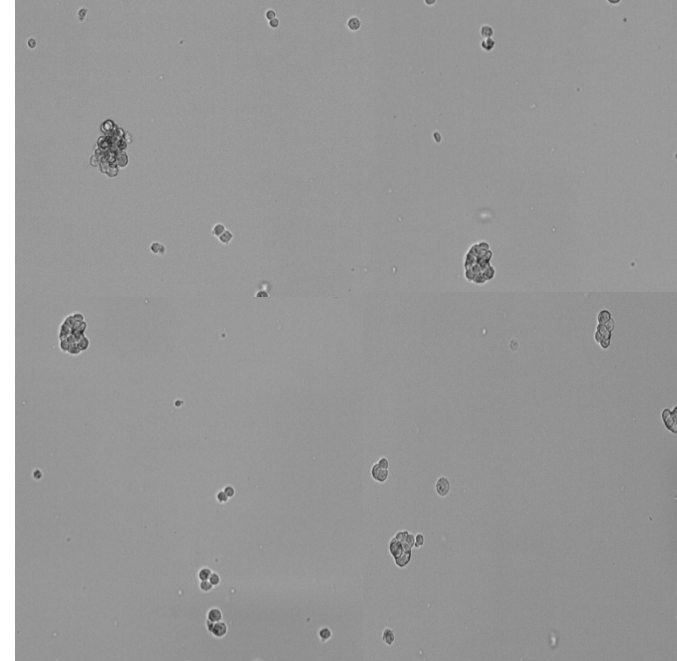




scRNA-seq requirements



Debris



Clumps



Conclusions

- Sequencing: short and long reads serve **different purposes**
- Most technologies rely on **living cells**
- Many **new technologies** emerging: choose your technology wisely
- Major technology providers: **10x Genomics, BD, Parse**
- Define your **research question** well
- Most crucial parameters: **viability, count, low-debris**

